

**Target-Conditioned Party Entity Detection
in Tamil Nadu Political**

A project work submitted to,
St. Joseph's College (Autonomous)
Affiliated to Bharathidasan University, Tiruchirappalli

in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF SCIENCE IN STATISTICS

Submitted by

BRAXTON GENO ANAND B	23UST105
THARUNKUMAR M	23UST108
MOHAMED YASAR ARAFADH A	23UST135
MYSTICA ROSE D	23UST156
AARTHI S	23UST157

Under the guidance of

Dr. Lilly George, M.Sc., M.Phil., Ph.D.
Assistant Professor of Statistics



Department of Statistics
St. Joseph's College (Autonomous)
Accredited at A⁺⁺ Grade (Cycle-IV) by NAAC
Special Heritage Status Awarded by UGC
Tiruchirappalli - 620 002

APRIL – 2026

Dr. LILLY GEORGE
Assistant Professor
Department of Statistics
St. Joseph's College (Autonomous)
Tiruchirappalli - 620 002



CERTIFICATE

This is to certify that the project entitled “**Target-Conditioned Party Entity Detection in Tamil Nadu Political Tweets**”, submitted to the Department of Statistics, St. Joseph's college (Autonomous), Tiruchirappalli – 620 002, in partial fulfillment for the award of the degree of **Bachelor of Science in Statistics**, is an original work done by by **Mr. Braxton Geno Ananad B (23UST105), Mr. Tharunkumar M (23UST108), Mr. Mohamed Yasar Arafadh A (23UST135), Ms. Mystica Rose D (23UST156), Ms. Aarthi S (23UST157)**, under my supervision and guidance during the academic period November 2025 to March 2026.

Signature of the HOD

Dr. R. Vijayakumar

Assistant Professor and Head

Department of Statistics

St. Joseph's College (Autonomous)

Tiruchirappalli-620 002

Signature of the Guide

Dr. Lilly George

Assistant Professor

Department of Statistics

St. Joseph's College (Autonomous)

Tiruchirappalli-620 002

Submitted for the viva-voce examination on _____

Signature of the External Examiner

Date:

Signature of the Internal Examiner

Date:

CERTIFICATE OF SUSTAINABLE DEVELOPMENT GOALS (SDG) COMPLIANCE

This is to certify that the project work entitled has been “**Target-Conditioned Party Entity Detection in Tamil Nadu Political Tweets**” done in alignment with the **United Nations Sustainable Development Goals (SDGs)** by **Mr. Braxton Geno Ananad B (23UST105), Mr. Tharunkumar M (23UST108), Mr. Mohamed Yasar Arafadh A (23UST135), Ms. Mystica Rose D (23UST156), Ms. Aarthi S (23UST157)**, the students of the Department of Statistics, St. Joseph’s College (Autonomous), Tiruchirappalli – 620 002, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Science in Statistics**.

The project is aligned with **SDG 16 – Peace, Justice and Strong Institutions**, focuses on promoting peace, justice, and strong institutions. It aims to reduce violence, ensure equal access to justice for all, protect human rights, and build transparent, accountable, and effective institutions that serve society fairly.

This project was carried out under my supervision and guidance during the academic period **November 2025 to March 2026**. The work is found to be original and reflects the student’s ability to apply sustainability principles in statistical fields.

Signature of the HOD

Dr. R. Vijayakumar

Assistant Professor and Head

Department of Statistics

St. Joseph’s College (Autonomous)

Tiruchirappalli-620 002

Signature of the Guide

Dr. Lilly George

Assistant Professor

Department of Statistics

St. Joseph’s College (Autonomous)

Tiruchirappalli-620 002

Submitted for the viva-voce examination on _____

Signature of the External Examiner

Date:

Signature of the Internal Examiner

Date:

Braxton Geno Anand B (23UST105)
Tharunkumar M (23UST108)
Mohamed Yasar Arafadh A (23UST135)
Mystica Rose D (23UST156)
Aarthi S (23UST157)

III B.Sc. Statistics

Department of Statistics

St. Joseph's College (Autonomous)

Tiruchirappalli - 620 002



DECLARATION

We hereby declare that the project work entitled Target-Conditioned Party Entity Detection in Tamil Nadu Political Tweets is a record of original work done by **Mr. Braxton Geno Ananad B (23UST105), Mr. Tharunkumar M (23UST108), Mr. Mohamed Yasar Arafadh A (23UST135), Ms. Mystica Rose D (23UST156), Ms. Aarthi S (23UST157)** during the period of study, under the guidance of **Dr. Lilly George**, Assistant Professor, Department of Statistics, St. Joseph's College, Tiruchirappalli – 620 002.

We further declare that any part of this project work has not been submitted elsewhere for the award of any Degree or Diploma at any other University or Research Institute.

Signature of the Candidates

ACKNOWLEDGEMENT

First and foremost, we would like to express our thanks to **GOD** for his love and strength in helping us to finish the project.

We wish to extend our sincere thanks to **Rev. Dr PAVULRAJ MICHAEL SJ**, Rector, St. Joseph's College (Autonomous), Tiruchirappalli.

We wish to extend our wholehearted thanks to **Rev. Dr M. AROCKIASAMY XAVIER SJ**, Secretary, St. Joseph's College (Autonomous), Tiruchirappalli.

We wish to thank **Rev. Dr S. MARIADOSS SJ**, Principal, St. Joseph's College (Autonomous), Tiruchirappalli.

We express our deep sense of gratitude and indebtedness to **Dr R. VIJAYAKUMAR, M.Sc., M.Phil., PhD**, Head of the Department of Statistics, for encouraging us.

We hereby express our profound thanks to our beloved guide, **Dr. LILLY GEORGE, M.Sc., M.Phil., PhD**, Assistant Professor of Statistics, for guiding us and for all the help extended to us throughout the completion of our project work.

We also convey our thanks to **Dr. J. GLORYPERSIAL M.Sc., M.Phil., MBA., PGDCA., Ph.D., Dr. T. VENKATESAN, M.Sc., M.Sc., M.Phil., Ph.D., and Dr. A. PHILIP AROKIADOSS, M.Sc., M.Phil., PGDCA., Ph.D.**, Assistant Professors, Department of Statistics, St. Joseph's College (Autonomous), Tiruchirappalli, for their encouragement and support.

Finally, we convey our thanks to our family members and friends for supporting us in our project.

CONTENT

1	CHAPTER 1 Introduction	1
2	CHAPTER 2 Review of Literature	19
3	CHAPTER 3 Research Methodology	27
4	CHAPTER 4 Statistical Analysis	55
5	CHAPTER 5 Conclusion	65
6	BIBLIOGRAPHY	69
7	APPENDIX	72

CHAPTER 1

INTRODUCTION

INTRODUCTION

Background: Political Discourse in Tamil Nadu's Digital Ecosystem

The rapid proliferation of social media platforms, specifically **X (formerly Twitter)**, has fundamentally transformed the landscape of political communication across India. In the specific context of **Tamil Nadu**, a state characterized by exceptionally high levels of political engagement and a deeply rooted, unique linguistic identity, these digital platforms have emerged as a primary arena for public discourse, policy debate, and electoral mobilization. For many citizens and political stakeholders, the digital ecosystem is now the front line where political narratives are shaped and contested in real-time.

The Rise of Social Media Political Discourse

Digital platforms have democratized political participation in Tamil Nadu, allowing for immediate interaction between leadership and the electorate. However, the sheer volume of data generated on these platforms necessitates advanced computational tools to monitor and understand the shifting tides of public opinion. Traditional methods of manual monitoring are no longer scalable given the velocity and scale of "X" data in the regional political sphere.

Tamil Linguistic Complexity and Computational Challenges

While these platforms offer a wealth of information, the computational analysis of Tamil political text presents significant technical hurdles. Tamil is a **morphologically rich and agglutinative** language, meaning a single root word can take hundreds of different forms through the addition of suffixes. This complexity often defeats traditional Natural Language Processing (NLP) tools that rely on simple keyword matching. Furthermore, the digital ecosystem in Tamil Nadu is characterized by:

- **Code-switching:** The frequent and fluid mixing of Tamil and English (often referred to as Tanglish) within a single post.
- **Nuanced Rhetoric:** The pervasive use of irony, metaphors, and sarcasm, which are hallmark features of Tamil political wit but are notoriously difficult for machines to interpret.
- **Linguistic Noise:** The presence of URLs, emojis, and informal grammar that requires specialized preprocessing

Regional Importance and the Research Gap

Accurately identifying mentions of political entities—such as **ADMK, BJP, Congress, DMK, NTK, and TVK**—is the foundational step for any high-level analysis, such as sentiment monitoring or policy feedback loops. Standard Named Entity Recognition (NER) tasks often struggle to determine if a party is the actual subject of a discussion or merely a peripheral mention.

Most existing research in the Tamil domain has focused on general multiclass sentiment analysis (labelling a tweet as Positive, Negative, or Neutral). There is a critical gap in regional language technologies for **Target-Conditioned Party Entity Detection (TC-PED)**—a method that doesn't just look at the tweet in isolation but validates the relevance of the text in relation to a specific candidate party. This study addresses this gap by leveraging **IndicBERT-v2**, a transformer model optimized for Indian scripts, to provide a sophisticated and scalable methodology for regional political monitoring.

Problem Statement: Entity-Specific Political Detection in Tamil Tweets

The primary challenge addressed in this study is the inability of traditional computational methods to accurately identify and validate specific political entity mentions within the complex environment of Tamil social media. While standard Natural Language Processing (NLP) techniques have evolved, they remain insufficient for the specific demands of regional political monitoring in Tamil Nadu.

The Failure of Traditional NER and Multiclass Classification

Traditional **Named Entity Recognition (NER)** and standard multiclass classification models encounter significant roadblocks when applied to political discourse on platforms like X (Twitter):

- **Contextual Ambiguity:** Standard NER often struggles to distinguish whether a political party is the actual subject of a discussion or merely a peripheral mention.
- **Treating Tweets as Single Units:** Standard BERT-based classification often treats an entire tweet as a single unit, which frequently results in missing specific target parties when multiple entities are mentioned within the same post.
- **Morphological Richness:** Tamil is an agglutinative language where a single root word (like a party name) can take hundreds of forms through case markings and suffixes. Traditional models often fail to recognize these variations as belonging to the same

entity.

- **Linguistic Noise:** The frequent use of code-switching (Tamil-English), irony, and metaphors in political rhetoric creates a "noisy" environment that linear models and keyword-matching systems cannot navigate effectively.

Identifying the Research Gap

Current research in Tamil NLP primarily focuses on general multiclass sentiment analysis—categorizing tweets broadly as Positive, Negative, or Neutral. However, in a complex political landscape where a single tweet might praise one party while simultaneously criticizing another, general sentiment scores lose their utility.

There is a distinct lack of frameworks that apply **Target-Conditioned Party Entity Detection (TC-PED)**. Existing systems rarely utilize relational bridges, such as the [CLS] token, to specifically validate the link between the text's deep semantic meaning and a specific party label. This project fills this gap by moving beyond simple keyword identification toward a sophisticated model that performs **entity-specific validation**, ensuring that the detection is conditioned on the target party rather than a general classification of the text.

Proposed Framework: Target-Conditioned Party Entity Detection (TC-PED)

To address the limitations of traditional classification, this study proposes the **Target-Conditioned Party Entity Detection (TC-PED)** framework. Conceptually, the model shifts from a general multi-label approach to a specialized relational validation system, designed to determine if a specific political party is the actual subject of a given tweet.

Binary Relevance-Inspired Problem Transformation

The framework begins with a fundamental restructuring of the data. Traditional models attempt to classify a single tweet into multiple categories simultaneously, which often dilutes the context. Our framework utilizes a **Binary Relevance-inspired transformation**.

- **Target-Tweet Pairs:** Each raw tweet is paired individually with every candidate party (ADMK, BJP, Congress, DMK, NTK, and TVK).
- **Data Expansion:** This transforms the original dataset of N tweets into a larger dataset of $N \times 6$ samples.
- **Binary Objective:** Each pair is assigned a binary label ($\in \{0,1\}$), indicating whether

that specific target party is mentioned within that specific context.

CLS Relational Embedding and Target Conditioning

The core of the TC-PED model lies in its ability to create a deep semantic bridge between the text and the entity.

Input Formatting: We condition the model by passing the input as [CLS] Tweet Text [SEP] Candidate party [SEP]

- **Relational Bridge:** By passing the party name as a secondary sequence, the transformer's self-attention mechanism allows every word in the tweet to "attend" specifically to the candidate party.
- **CLS Token as Aggregator:** The model extracts **the [CLS] (Classification) token embedding**—a 768-dimensional vector ($h_{CLS} \in \mathbb{R}^{768}$). Unlike standard embeddings, this specific vector represents the aggregate semantic relationship and relational "weight" between the tweet and the specific party target.

By combining this target-specific conditioning with the linguistic power of **IndicBERT-v2**, the TC-PED framework moves beyond simple keyword matching to a sophisticated model capable of high-precision entity detection in imbalanced, real-world social media settings.

Importance of Target-Conditioned Models in Political NLP

In the volatile landscape of digital political discourse, general classification models often fall short because they fail to account for the entity-specific nuances of language. The transition to Target-Conditioned Party Entity Detection (TC-PED) represents a shift toward higher granularity and statistical reliability in regional language processing.

Why General Sentiment is Insufficient

Most existing research in Tamil Natural Language Processing (NLP) is limited to general multiclass sentiment analysis, which categorizes a tweet as simply "Positive," "Negative," or "Neutral". In a political context, this approach is fundamentally flawed because:

- **Subjectivity Blindness:** A tweet may contain strong negative sentiment directed at an issue (e.g., inflation) while maintaining a positive or neutral stance toward a specific party.
- **Lack of Attribution:** General sentiment models tell us how a person feels but often fail to clarify who those feelings are directed toward, rendering the data less actionable

for political stakeholders.

Need for Entity-Specific Validation

Political entities in Tamil Nadu, such as the DMK, ADMK, or BJP, are often discussed using complex linguistic structures, including irony and metaphors.

- **Relational Context:** TC-PED moves beyond keyword matching by using the [CLS] token as a "relational bridge" between the tweet text and a specific party label.
- **Morphological Awareness:** Since Tamil is an agglutinative language where party names frequently change their suffixes (case markings), entity-specific validation ensures the model captures the "root" entity regardless of its grammatical form.

Handling Multi-Party Mentions

A significant limitation of standard BERT-based classification is that it treats a tweet as a single unit, which often results in the model missing specific targets when multiple parties are mentioned in one post.

- **Contextual Overlap:** In a landscape where one tweet might praise one party while criticizing another, target-conditioned models allow the self-attention mechanism to focus specifically on the party in question.
- **One-vs-Rest Success:** By transforming the problem into "Target-Tweet" pairs, the model can maintain high precision for each party individually, even when they appear together in the same raw text.

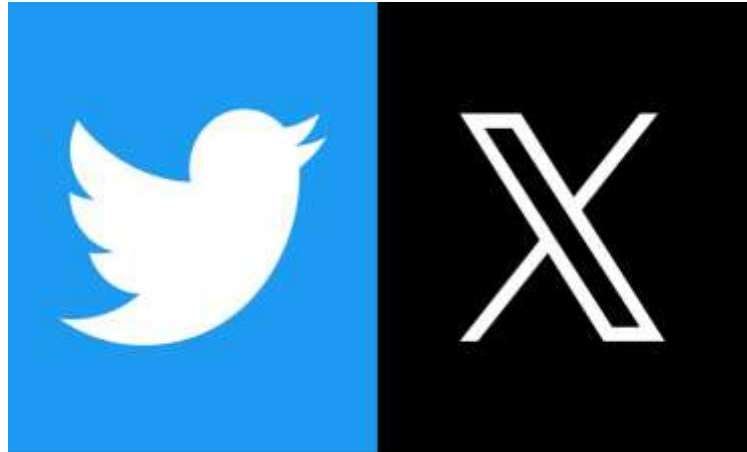
Real-Time Monitoring Applications

The ability to perform precise, target-conditioned detection opens the door for scalable and reliable monitoring tools.

- **Data-Driven Insights:** This framework provides a methodology for real-time political sentiment and entity monitoring in regional languages, allowing for a more accurate reflection of the "ground reality" on social media.
- **Adaptive Thresholds:** With a high ROC-AUC (0.8780), these models can be adjusted to favour precision or recall depending on the specific monitoring needs, such as tracking electoral mobilization or identifying emerging policy feedback.

Why X (Formerly Twitter) as the Primary Data Source

The choice of **X (formerly Twitter)** as the primary data source for this study is driven by its unique role as the digital town square for political discourse. Unlike other social media platforms, X offers specific structural and cultural advantages that make it an ideal environment for studying real-time political entity detection in Tamil Nadu



Political Immediacy

In the modern political landscape, X has fundamentally transformed communication by providing a space for instantaneous updates and reactions.

- **Real-Time Arena:** It serves as a primary arena for electoral mobilization and policy debate where narratives are formed and contested in seconds.
- **Temporal Dynamics:** The "vocabulary of conflict" and support on the platform is highly dynamic, reflecting the most current political developments as they happen.

Public Discourse Visibility

The platform provides an unparalleled window into the public's linguistic identity and political engagement.

- **Linguistic Diversity:** It captures the authentic use of the Tamil language in a digital context, including complex code-switching and the use of regional metaphors.
- **Open Access:** Unlike private messaging apps, the public nature of X discourse allows for the curation of manually annotated datasets that reflect real-world political sentiment.

Elite + Grassroots Presence

X is unique in its ability to bring diverse political stakeholders into the same digital ecosystem.

- **Top-Down and Bottom-Up:** The platform hosts everyone from high-level political leaders to grassroots activists, making it possible to identify mentions of major stakeholders like **ADMK, BJP, Congress, DMK, NTK, and TVK** across all levels of society.
- **High Engagement:** Tamil Nadu is characterized by exceptionally high political engagement on the platform, ensuring a rich data stream for entity-specific monitoring.

Short-Form Argumentative Structure

The character-limited nature of X forces users into a condensed, often argumentative style of communication that is particularly useful for training detection models.

- **Agglutinative Challenges:** The short-form text highlights the challenges of Tamil's SOV (Subject-Object-Verb) structure and its morphologically rich nature, where root words change suffix based on the context of the argument.
- **Dense Context:** Because users must be concise, tweets often contain a high density of entity mentions, irony, and sarcasm, providing a rigorous test for the TC-PED framework's ability to capture deep semantic relationships.

Selection Criteria for Political Parties

The effectiveness of a **Target-Conditioned Party Entity Detection (TC-PED)** system depends heavily on the selection of representative entities that reflect the actual political landscape. For this study, six major political parties—**ADMK, BJP, Congress, DMK, NTK, and TVK**—were selected as the target entities. The inclusion of these specific parties is justified by their electoral relevance, ideological diversity, and significant digital footprint within the Tamil Nadu ecosystem.

Electoral Relevance and Regional Stakeholders

The selected parties represent the primary stakeholders in Tamil Nadu's governance and opposition.

- **Established Powerhouses:** The DMK and ADMK are the traditional Dravidian pillars that have dominated the state's electoral history for decades.

- **National Presence:** The BJP and Congress represent the major national-level parties that play a significant role in coalition politics and regional discourse.
- **Emerging and Identity Politics:** The inclusion of the NTK and the newly formed TVK reflects the evolving nature of the political landscape, capturing the shift toward newer regional alternatives.

Ideological Diversity

To ensure the model is robust enough to handle various "vocabularies of conflict" and support, the selection spans a wide ideological spectrum.

- **Dravidianism vs. Nationalism:** The dataset captures the linguistic nuances of the debate between regional Dravidian identity and nationalistic narratives.
- **Varied Rhetoric:** By including parties with distinct ideological stances, the model is trained to recognize a broader range of metaphors, ironies, and political slogans unique to each group.

Digital Presence and Data Density

As this study utilizes **X (formerly Twitter)** as its primary source, the parties were chosen based on their high levels of digital engagement.

- **High Engagement:** These six parties are the most frequently discussed entities on Tamil social media, ensuring sufficient data for manual annotation and model training.
- **Real-World Imbalance:** Selecting these parties allows the study to test the model under real-world conditions where certain parties may have a significantly higher "volume" of mentions than others, creating a natural class imbalance that the **MCC** metric is designed to evaluate.

Overview of Selected Political Parties

The selection of target entities for the **TC-PED** framework is based on their foundational role in the regional political ecosystem. These parties represent the primary stakeholders whose mentions the model is designed to validate within the "Target-Tweet" binary transformation.

1. Dravida Munnetra Kazhagam (DMK)

- **Party Name:** Dravida Munnetra Kazhagam (DMK).
- **Founded:** 1949.

- **Ideology:** Dravidianism, Social Justice, and State Autonomy.

Current Relevance in Governance

The DMK is a cornerstone of Tamil Nadu's political identity, currently serving as the ruling party in the state as of February 2026. Its governance model emphasizes the "Dravidian Model," focusing on inclusive growth, linguistic pride, and federalism. As the party in power, it is a primary subject of both administrative praise and opposition critique on social media, making it a high-frequency target for entity detection.

Digital Political Engagement Patterns

On platforms like X (formerly Twitter), the DMK maintains a highly organized digital presence.

- **Narrative Control:** The party utilizes social media to broadcast policy updates and counter-narratives from opposition parties like the ADMK and BJP.
- **Linguistic Nuance:** DMK-related discourse often features high-register Tamil and classical literary references, which tests the **IndicBERT-v2** model's ability to handle morphologically rich text.
- **Code-Mixed Discourse:** Supporters and critics alike frequently use "Tanglish" (Tamil-English code-switching) when discussing DMK policies, necessitating the semantic retention and normalization steps in the TC-PED preprocessing pipeline.
- **Target Dynamics:** In the TC-PED framework, DMK mentions are often embedded in complex "one-vs-rest" scenarios where a single tweet might mention the DMK alongside its rivals, requiring the model to use the **[CLS] token** to establish specific relational relevance.

2.All India Anna Dravida Munnetra Kazhagam (ADMK)

- **Party Name:** All India Anna Dravida Munnetra Kazhagam (AIADMK)
- **Founded:**1972
- **Ideology:** Populism, Welfarism, and Secularism (Annaism)

Dravidian Offshoot and Foundation

The AIADMK was founded by the legendary actor-politician **M. G. Ramachandran (MGR)** in 1972 following his expulsion from the DMK. It emerged as a "Dravidian offshoot" that successfully blended the core tenets of the Dravidian movement—such as social justice

and linguistic pride—with a powerful brand of charismatic populism. Under the leadership of MGR and later **J. Jayalalithaa**, the party cultivated a massive, loyalist base among the rural poor and women, establishing itself as a dominant force in Tamil Nadu's bipolar political system.

Electoral Competition with DMK

The AIADMK has spent much of its history in direct, fierce competition with the DMK. This rivalry has defined Tamil Nadu's politics for over five decades.

- **The Bipolar Axis:** The state's electoral history is essentially a see-saw battle between these two "poles." As of February 2026, the AIADMK serves as the principal opposition, leading a robust coalition (including the BJP and PMK) to challenge the DMK's incumbency.
- **Competitive Populism:** The competition often manifests through welfare schemes. While the DMK focuses on systemic reforms like the "Dravidian Model," the AIADMK has historically prioritized direct social benefits, such as the midday meal scheme and free laptop initiatives, which are central themes in the party's social media discourse.

Social Media Strategy Differences

Compared to the DMK's more centralized and high-register linguistic approach, the AIADMK's digital strategy often leans toward **grassroots-oriented and aggressive counter-narratives**.

- **"Urutukalum Thirtukalum" (Lies and Deceit):** In recent campaign cycles, the AIADMK has utilized targeted digital campaigns to critique the ruling governments "failed" promises. Their strategy focuses on highlighting micro-level issues, such as the state of the economy and agricultural growth.
- **Target-Conditioned Sensitivity:** On X (formerly Twitter), AIADMK discourse is heavily marked by references to "Amma" (Jayalalithaa) and "Puratchi Thalaivar" (MGR). For the TC-PED framework, this creates a specific set of **semantic markers**. A tweet might not mention "AIADMK" explicitly but use these honorifics; the model's ability to link these terms to the party entity through **relational embeddings** is crucial for accurate detection.
- **Allied Echoes:** As the lead of the National Democratic Alliance (NDA) in the state, AIADMK's digital presence frequently overlaps with BJP narratives, requiring the

model to distinguish between the party's core regional identity and its broader national alliances.

3. Bharatiya Janata Party (BJP)

- **Party Name:** Bharatiya Janata Party (BJP)
- **Founded:** 1980 (successor to the Bharatiya Jana Sangh)
- **Ideology:** Hindutva (Nationalism), Integral Humanism, Conservatism

National Ruling Party and Contextual Relevance

As the dominant national ruling party under the leadership of Prime Minister Narendra Modi, the BJP carries significant weight in Tamil Nadu's political discourse. While historically a smaller player in the state compared to the Dravidian giants, its status as the central authority makes it a constant focal point for debates regarding federalism, tax devolution, and national integration. In the digital ecosystem, the BJP is often positioned as the ideological antithesis to the Dravidian "rationalist" tradition, creating a high-contrast environment for textual analysis.

Expanding Footprint in Tamil Nadu

Under the state leadership of **K. Annamalai**, a former IPS officer, the BJP has aggressively sought to position itself as a viable alternative to the DMK and AIADMK.

- **The "Third Pole" Ambition:** Moving away from being a mere "junior partner" in alliances, the BJP significantly increased its vote share in the 2024 Lok Sabha elections.
- **Electoral Strategy for 2026:** As of early 2026, the party has re-established its alliance with the AIADMK under the National Democratic Alliance (NDA) banner. This strategic realignment aims to consolidate the anti-incumbency vote against the DMK, making "BJP" a critical target entity for detection in coalition-related social media posts.

Strong Centralized Digital Campaign Ecosystem

The BJP is widely recognized for having one of the most sophisticated digital campaign infrastructures in India, and its Tamil Nadu unit is no exception.

- **Annamalai's Digital Impact:** The party leader's "En Mann En Makkal" (My Land, My People) padayatra was heavily live-streamed and promoted on X, generating a massive volume of Tamil-language data.

- **Standardized Vocabulary:** The party's digital presence often uses specific slogans (e.g., "**Thamarai Malarum**" – The Lotus will bloom) and high-quality multimedia that trigger distinct engagement patterns.
- **Targeting Challenges:** For the TC-PED framework, the BJP presents a unique challenge: it is often mentioned in "national vs. regional" debates. The model must distinguish when "BJP" is being discussed in a local Tamil Nadu administrative context versus broad national policy, using the [CLS] **relational embedding** to ensure the entity detection remains contextually grounded.

4. Indian National Congress (INC)

- **Party Name:** Indian National Congress (INC)
- **Founded:** 1885
- **Ideology:** Social Liberalism, Secularism, and Inclusive Development

Historical National Party

The Indian National Congress (INC) holds a unique position as the oldest political organization in India. While it was once the dominant force in Tamil Nadu's governance (the "Kamaraj era"), it has transitioned over the last few decades into a vital ideological and coalition partner within the state. In the context of Tamil political discourse, the Congress represents a bridge between national secular values and regional aspirations, often acting as a stabilizing force within the broader "Secular Progressive Alliance" (SPA).

Alliance Politics in Tamil Nadu

The Congress's relevance in Tamil Nadu is deeply intertwined with its strategic partnership with the DMK.

- **The SPA Coalition:** As of early 2026, the Congress remains a core constituent of the DMK-led alliance. This partnership is built on shared ideological opposition to the BJP's national policies.
- **Electoral Synergy:** The party maintains a consistent vote share and holds significant influence in specific southern and western pockets of the state. Its presence on social media often mirrors the collective messaging of the alliance, making it a critical entity for the TC-PED framework to identify, particularly in posts discussing coalition seat-sharing and joint campaign rallies.

Digital Narrative Positioning

The digital footprint of the Congress in Tamil Nadu is characterized by a blend of national leadership focus and local issue advocacy.

- **National-Regional Sync:** The party's social media strategy frequently amplifies the "Bharat Jodo" (Unite India) ethos, adapting it to local Tamil sentiments regarding linguistic rights and social justice.
- **The "Anti-Establishment" Voice:** On X (formerly Twitter), the Congress TN unit is highly active in critiquing the central government's economic policies (GST, inflation) while supporting the DMK's state-level initiatives.
- **Detection Challenges:** For the TC-PED model, the Congress presents a "Relational Overlap" challenge. Because it is almost always mentioned alongside the DMK in the current political climate, the **binary relevance transformation** is essential. The model must learn to distinguish between a tweet that is "About the DMK but mentions Congress" versus a tweet where the "Congress" is the primary subject of the political validation.

5. Naam Tamilar Katchi (NTK)

- **Party Name:** Naam Tamilar Katchi (NTK)
- **Founded:** 2010 (Reorganized by Seeman)
- **Ideology:** Tamil Nationalism (Tamil Desiyam), Environmentalism, and Social Justice

Regional Nationalist Ideology

The Naam Tamilar Katchi (NTK) represents a significant departure from the traditional Dravidian identity politics of the DMK and AIADMK. Under the leadership of **Senthamilan Seeman**, the party advocates for **Tamil Nationalism**, which prioritizes linguistic identity over broader regional or national identities. Its "Tamil for Tamils" stance rejects the Dravidian label as an outsider construct, calling for a "Green Revolution" and a "Cultural Revolution" that places ethnic Tamils at the center of governance. This ideological purity has carved out a unique space in the state's political landscape, appealing to those disillusioned with the established "bipolar" system.

Youth Digital Mobilization

NTK has distinguished itself as a digital-first organization, leveraging social media to bypass traditional media gatekeepers.

- **Volunteer-Led Ecosystem:** The party relies on a vast, decentralized network of young volunteers known as "Thambis" (younger brothers) and "Thangais" (younger sisters), who dominate X (formerly Twitter) with coordinated hashtags and aggressive defense of the party's stance.
- **Educational Outreach:** The party's digital wings often produce long-form explanatory videos and historical revisionist content designed to educate the younger generation on "pure" Tamil history and the perceived failures of national and Dravidian parties.
- **Target-Conditioned Relevance:** In the context of our study, the NTK's digital presence is characterized by high-frequency mentions that are often detached from mainstream alliance discourse, as the party consistently contests elections alone (Solo Contest strategy).

High Engagement Online Rhetoric

The rhetoric of the NTK is tailor-made for the attention economy of social media.

- **Seeman's Oratory:** Seeman is arguably the most viral political orator in Tamil Nadu. Snippets of his speeches—marked by high-register Tamil, emotional appeals, and provocative critiques—are widely shared, creating a dense stream of textual data for detection.
- **The "Third Force" Narrative:** Online discussions surrounding the NTK often focus on its rapid growth in vote share (surpassing 6% in the 2021 Assembly elections and performing as the "third-largest party" in many pockets in 2024). This creates a specific "vocabulary of growth" that the **TC-PED framework** must differentiate from the established power of the DMK and AIADMK.
- **Detection Sensitivity:** Because NTK supporters are highly active in "tagging" and mentioning the party in response to every major socio-political event, the model must be sensitive to distinguishing between organic support and the coordinated digital activism that characterizes the NTK's online footprint.

6. Tamilaga Vettri Kazhagam (TVK)

- **Party Name:** Tamilaga Vettri Kazhagam (TVK)
- **Founded:** February 2, 2024
- **Ideology:** Secular Social Justice, Egalitarianism, and Transparency (Ambedkar-Periyar-Kamaraj path)

Emerging Political Formation

Tamilaga Vettri Kazhagam (TVK) is the newest major entrant in Tamil Nadu's political landscape, founded by actor-turned-politician **Vijay** (Joseph Vijay Chandrasekhar). Launched with the explicit goal of contesting the 2026 Assembly elections, the TVK seeks to position itself as a "centrist-alternative" to the long-standing Dravidian duopoly. By adopting the "Whistle" symbol and focusing on a corruption-free, accountable governance model, the party aims to capture the "neutral" and "anti-incumbency" votes that have historically fluctuated between the DMK and AIADMK.

Celebrity-Driven Mobilization

The foundation of TVK's organizational strength lies in the Vijay Makkal Iyakkam (VMI), a massive fan-club network that has been active in social welfare for over a decade.

- **Fanbase to Cadre:** Unlike traditional parties that build from grassroots ideological cells, TVK has successfully converted a sprawling celebrity fanbase into a structured political cadre. This transition was showcased during the party's inaugural conference in Vikravandi (October 2024), which reportedly drew over 800,000 attendees.
- **Leadership Iconography:** The party relies heavily on Vijay's "mass appeal" and clean public image. His speeches often address the youth directly, framing the political plunge as a duty to the "soil and the people" (*En Mann En Makkal* sentiment), which resonates deeply with the state's cinema-integrated political culture.

Strong Early Digital Footprint

TVK has demonstrated a sophisticated understanding of the digital ecosystem, often outperforming established parties in viral engagement.

- **Virtual Warriors:** Vijay has famously referred to his IT and social media wing as the "best in India," tasking them to function as "virtual warriors" who maintain "decency and dignity" while defending the party's narrative.
- **Global Diaspora Outreach:** The party has appointed social media coordinators not just across Tamil Nadu, but also in other Indian states and countries with significant Tamil populations (USA, UK, Malaysia, etc.), creating a globalized echo chamber for its campaign.
- **AI and Innovation:** TVK has embraced advanced technology, including the use of AI-generated videos of historical icons like C.N. Annadurai to establish ideological

continuity. For the TC-PED framework, this high-volume, tech-savvy digital presence produces a unique stream of data where party mentions are often tied to specific hashtags (#TVK, #ThalapathyVijay) and viral multimedia content, requiring the model to distinguish between "fan activity" and "political discourse."

Research Objectives

The primary goal of this study is to move beyond generic text classification and develop a specialized framework capable of understanding the nuances of Tamil political discourse. The study is guided by the following core objectives:

1) Develop a Target-Conditioned Detection Pipeline

The central objective is to design and implement a robust **Target-Conditioned Party Entity Detection (TC-PED)** pipeline specifically for the Tamil language.

- **Relational Mapping:** Unlike standard models that treat a tweet as an isolated unit, this objective focuses on creating a "relational bridge" between the tweet text and specific political entities (ADMK, BJP, Congress, DMK, NTK, and TVK).
- **Semantic Extraction:** The project aims to utilize the **[CLS] token embeddings** from **IndicBERT-v2** to capture deep semantic relevance, ensuring that the model distinguishes between a casual keyword mention and a meaningful discussion of a political party.
- **Architecture Innovation:** To transform a multi-label problem into a binary relevance task, allowing for more precise validation of each party entity.

2) Evaluate Performance Under Class Imbalance

In the real-world digital ecosystem of Tamil Nadu, political mentions are naturally skewed; certain parties dominate the discourse while others have a smaller, more concentrated footprint.

- **Metric Reliability:** A key objective is to evaluate the model using statistically rigorous metrics such as the **Matthews Correlation Coefficient (MCC)** and **ROC-AUC**, rather than simple accuracy, which can be misleading in imbalanced datasets.
- **Loss Optimization:** The study explores the use of **Weighted Binary Cross-Entropy Loss** to ensure the model does not become biased toward the majority "negative" class (where a party is not mentioned), thereby maintaining high recall for minority party mentions.

3) Demonstrate Feasibility for Regional Language Monitoring

This study seeks to bridge the gap between advanced Transformer-based NLP and practical regional application.

- **Morphological Handling:** The objective is to demonstrate that pre-trained multilingual models like IndicBERT-v2 can effectively handle the **agglutinative nature** and **code-mixed (Tanglish)** patterns found in Tamil social media.
- **Scalability for Stakeholders:** By providing a high-performance detection framework, the study aims to prove that automated tools can offer reliable, real-time insights for political analysts, policymakers, and researchers monitoring the Tamil digital ecosystem.

CHAPTER 2

REVIEW Of LITERATURE

REVIEW OF LITERATURE

A literature review is an overview of the previously published works related to the topic. A good literature review can ensure that a proper question has been asked and a proper theoretical or research methodology has been chosen. Here, the following literature was reviewed.

Title: Multilingual Transformer Contextual Embedding Model for Political Tweets

Analysis

Author: Vaibhav Khatavkar, Sneha Petkar, Archana S. Vaidya

Year of publication: 2025

Objective:

The primary objective of this study was to develop an effective multilingual transformer-based model for analysing political discourse on Twitter. The research aimed to address the linguistic diversity and contextual complexity present in political tweets, particularly in multilingual settings. The authors sought to enhance contextual understanding and classification accuracy by leveraging transformer-based contextual embeddings capable of capturing semantic nuances across languages.

Methods:

The study employed a multilingual transformer architecture to generate contextual embeddings for political tweets. A dataset consisting of multilingual political content was collected and pre-processed to handle noise, hashtags, mentions, and informal expressions typical of social media text. The transformer model was fine-tuned for downstream classification tasks such as sentiment or stance detection. Performance was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. Comparative experiments were conducted against baseline machine learning and traditional embedding approaches.

Results:

The results demonstrated that multilingual transformer-based contextual embeddings significantly improved classification performance compared to traditional feature-based and monolingual approaches. The model effectively captured cross-lingual semantic patterns and contextual dependencies in political tweets. The findings highlighted the importance of

contextualised representations in handling multilingual and politically sensitive social media data.

Title: Synapse@DravidianLangTech 2025: Multiclass Political Sentiment Analysis in Tamil X (Twitter) Comments: Leveraging Feature Fusion of IndicBERTv2 and Lexical Representations

Author: Suriya Kp, Durai Singh K, Vishal A S, Sachin Kumar S

Year of publication: 2025

Objective:

The objective of this study was to develop a multiclass political sentiment analysis system for Tamil Twitter comments as part of the DravidianLangTech 2025 shared task. The research aimed to address challenges in low-resource language processing and political sentiment classification.

Methods:

The system utilised IndicBERTv2 contextual embeddings combined with handcrafted lexical features through a feature fusion strategy. Preprocessing steps included normalisation of Tamil text, handling code-mixing, and removal of noise. The fused features were fed into a classification model and evaluated using standard sentiment classification metrics.

Results:

The proposed system achieved competitive performance in the shared task, demonstrating the effectiveness of combining contextual transformer embeddings with lexical representations. The results highlighted the importance of hybrid approaches for improving sentiment classification in low-resource and morphologically rich languages like Tamil.

Title: Improving Named Entity Recognition for social media with Data Augmentation

Author: Wenzhong Liu and Xiaohui Cui

Year of publication: 2023

Objective:

The objective of this research was to enhance the performance of Named Entity Recognition systems on noisy social media text by employing data augmentation strategies. The study aimed to address data sparsity and variability issues that negatively impact model

generalisation.

Methods:

Various data augmentation techniques were applied to expand the training dataset, including synonym replacement, back-translation, contextual word substitution, and noise injection. A neural NER model was trained on both original and augmented datasets. Experimental evaluation was conducted using social media corpora, and performance was assessed using standard sequence labelling metrics.

Results:

The results showed that data augmentation significantly improved model robustness and generalisation capability. Enhanced F1-scores were observed compared to models trained on original datasets alone. The study confirmed that augmentation techniques are effective in mitigating noise-related challenges in social media NER tasks.

Title: A Survey on Deep Learning for Named Entity Recognition

Author: Jing Li, Aixin Sun, Jianglei Han & Chenliang Li

Year of publication: 2020

Objective:

The purpose of this study was to provide a comprehensive survey of deep learning approaches applied to Named Entity Recognition. The authors aimed to systematically analyse advancements in neural architectures, representation learning, datasets, and evaluation methodologies within the NER research domain.

Methods:

The paper conducted an extensive literature review of deep learning-based NER models, categorising them into various architectural paradigms such as CNNs, RNNs, LSTMs, CRFs, and transformer-based models. The survey examined different embedding techniques, including word embeddings and contextualised representations. Comparative analyses of benchmark datasets and evaluation metrics were also presented to identify trends and performance improvements.

Results:

The survey highlighted the rapid evolution of NER systems from feature-engineered

models to deep contextualised architectures. Transformer-based models were identified as achieving superior performance across multiple benchmarks. The study also identified ongoing challenges such as domain adaptation, low-resource languages, and noisy social media text, providing insights for future research directions.

Title: Named Entity Recognition for Social Media Texts with Semantic Augmentation

Author: Yuyang Nie, Yuanhe Tian, Xiang Wan, Yan Song & Bo Dai

Year of publication: 2020

Objective:

The objective of this research was to enhance Named Entity Recognition in social media texts by incorporating semantic augmentation techniques. The study focused on enriching contextual understanding to overcome noise and ambiguity.

Methods:

The authors integrated external semantic knowledge and augmentation strategies into neural NER frameworks. Semantic information from knowledge bases and contextual embeddings was used to supplement training data. Experimental evaluation was performed on social media datasets using standard NER metrics.

Results:

The proposed semantic augmentation approach improved entity recognition accuracy and robustness. The results indicated better handling of ambiguous and informal expressions commonly found in social media text.

Title: BERT: Pre-training of Deep Bidirectional Transformers for Language

Understanding

Author: Jacob Devlin, Ming-Wei Chang, Kenton Lee, Kristina Toutanova

Year of publication: 2019

Objective:

The objective of this study was to introduce BERT (Bidirectional Encoder Representations from Transformers), a deep pre-trained language representation model designed to improve performance across a wide range of Natural Language Processing (NLP) tasks. The authors aimed to overcome the limitations of unidirectional language models by

enabling deep bidirectional contextual understanding. The study sought to demonstrate that large-scale pre-training followed by task-specific fine-tuning could significantly enhance performance in tasks such as Named Entity Recognition, question answering, and sentence classification.

Methods:

BERT was pre-trained on massive unlabelled corpora, including Books Corpus and English Wikipedia, using two self-supervised objectives: Masked Language Modelling (MLM) and Next Sentence Prediction (NSP). The Transformer architecture enabled the model to capture bidirectional contextual relationships between words. After pre-training, the model was fine-tuned with minimal architectural modifications for various downstream tasks. For NER tasks specifically, token-level classification layers were added on top of the BERT encoder. The evaluation was conducted on multiple benchmark datasets, such as GLUE and SQuAD, to validate generalisation capabilities.

Results:

The experimental results demonstrated that BERT achieved state-of-the-art performance across numerous NLP benchmarks at the time of publication. Significant improvements were observed in Named Entity Recognition, question answering, and natural language inference tasks. The study established pre-training with deep bidirectional transformers as a dominant paradigm in NLP, leading to widespread adoption and further advancements in transformer-based models. BERT's success also inspired the development of several domain-specific and multilingual variants.

Title: Multimodal Named Entity Recognition for Short Social Media Posts

Author: Seungwhan Moon, Leonardo Neves and Vitor Carvalho

Year of publication: 2018

Objective:

The study aimed to improve Named Entity Recognition performance in short social media posts by incorporating multimodal information. The authors hypothesised that visual context from accompanying images could enhance entity disambiguation in short and ambiguous text.

Methods:

A multimodal neural network architecture was proposed, integrating textual features with visual features extracted from images using convolutional neural networks (CNNs). The combined representation was used for sequence labelling. The model was evaluated on a dataset of social media posts containing both text and images.

Results:

The multimodal approach outperformed text-only NER models, particularly in cases where the textual context was limited. Visual information contributed to improved entity recognition accuracy, demonstrating the value of multimodal learning for social media applications.

Title: Results of the WNUT2017 Shared Task on Novel and Emerging Entity**Recognition**

Author: Leon Derczynski, Eric Nichols, Marieke van Erp & Nut Limsopatham

Year of publication: 2017

Objective:

The primary objective of this study was to address the limitations of traditional Named Entity Recognition (NER) systems when applied to noisy and dynamic social media data. Specifically, the research focused on identifying novel and emerging entities that are not present in existing knowledge bases or training datasets. The study aimed to provide a standardised evaluation framework through a shared task, enabling researchers to assess the effectiveness of different approaches in handling unseen entities in real-world social media environments.

Methods:

The authors organised the WNUT 2017 shared task, which provided participants with annotated datasets collected from social media platforms. These datasets included emerging and previously unseen named entities to simulate real-world challenges. Various participating teams implemented machine learning and deep learning-based NER models, including feature-engineered systems. The systems were evaluated using standard performance metrics such as precision, recall, and F1-score. The comparative evaluation allowed the researchers to analyse strengths and weaknesses across different modelling strategies.

Results:

The results revealed that recognising novel and emerging entities remains significantly more challenging than traditional NER tasks performed on well-structured text. Overall performance scores were lower compared to conventional benchmarks, highlighting the impact of noisy language, spelling variations, and limited contextual information in social media text. Neural network-based models showed competitive performance; however, substantial performance gaps indicated the need for more robust domain adaptation techniques and improved handling of unseen entities.

Title: Neural Architectures for Named Entity Recognition

Author: Guillaume Lample, Miguel Ballesteros, Sandeep Subramanian, Kazuya Kawakami & Chris Dyer

Year of publication: 2016

Objective:

The objective of this research was to design a neural network architecture for Named Entity Recognition that minimises the need for extensive manual feature engineering. The study aimed to demonstrate that deep learning models, particularly recurrent neural networks, could automatically learn effective representations for sequence labelling tasks such as NER.

Methods:

The authors proposed a bidirectional Long Short-Term Memory (BiLSTM) network combined with a Conditional Random Field (CRF) layer for structured prediction. Character-level representations were incorporated using character-based embeddings to capture morphological information. The model was trained and evaluated on standard benchmark datasets such as CoNLL 2003. Performance was measured using precision, recall, and F1-score metrics, and comparisons were made with traditional feature-based CRF systems.

Results:

The proposed neural architecture achieved state-of-the-art performance on benchmark NER datasets at the time of publication. The integration of character-level embeddings significantly improved the recognition of rare and unseen words. The results demonstrated that neural architectures could outperform traditional methods without relying on handcrafted linguistic features.

CHAPTER 3

RESEARCH METHODOLOGY

RESEARCH METHODOLOGY

Chapter Overview

This chapter presents the comprehensive research methodology adopted for the development and evaluation of the proposed Target-Conditioned Party Entity Detection (TC-PED) framework. The methodology encompasses the complete experimental pipeline, beginning with large-scale data acquisition from X (formerly Twitter), followed by manual annotation, structured preprocessing, dataset transformation, model architecture design, training configuration, and systematic evaluation.

Data Collection and Acquisition

This section describes the data source, query formulation strategy, and scraping methodology adopted to construct the political tweet dataset used in this study. Wherever necessary, relevant code snippets are provided to ensure reproducibility and methodological transparency.

Data Source: X (Formerly Twitter)

The primary data source for this study is X (formerly Twitter), a widely used microblogging platform that facilitates real-time public discourse. X was selected for the following reasons:

- It enables large-scale political communication and public engagement.
- It supports short-form argumentative content suitable for entity detection tasks.
- Political discussions frequently exhibit measurable engagement (likes, retweets, replies), which can be used as a quality filter.

To retrieve politically relevant tweets related to Tamil Nadu, an advanced Boolean query was constructed. The query incorporated:

- Political keywords in Tamil and English,
- Party names and abbreviations,
- Leader names,
- Hashtags,
- Engagement filters,
- Language constraints.

The complete query used for data extraction is shown below:

```
QUERY = '("Tamil Nadu" OR TamilNadu OR தமிழ்நாடு OR தமிழகம்  
OR #TamilNadu OR #TNPolitics OR #TNPolls OR DMK OR AIADMK  
OR திமுக OR அதிமுக OR "நாம் தமிழர்" OR பாஜக OR TVK OR தவெக  
OR #DMK OR #AIADMK OR Stalin OR Annamalai OR EPS OR Vijay  
OR விஜய்) (lang:ta OR lang:en) min_faves:50  
min_retweets:20 filter:has_engagement -filter:replies'
```

The engagement constraints:

- min_faves:50
- min_retweets:20
- filter:has_engagement

were applied to ensure that only tweets with significant public interaction were included. Additionally, replies were excluded to focus on primary discourse rather than conversational threads.

Data Scraping Procedure

Data scraping was implemented using the twikit Python library. Tweets were collected from two categories:

- 200 Latest tweets
- 200 Top tweets

After merging both sets and removing duplicates, a total of 376 unique tweets were retained.

The core scraping loop with pagination and rate-limit control is shown below:

```
tweets = await client.search_tweet(QUERY, product='Top')
```

```
while count < max_tweets:
```

```
    for tweet in tweets:
```

```
        tweets_data.append({  
            'tweet_id': tweet.id,  
            'created_at': tweet.created_at,  
            'full_text': tweet.full_text,  
            'lang': tweet.lang,  
            'retweet_count': tweet.retweet_count,  
            'favorite_count': tweet.favorite_count,
```

```

        'reply_count': tweet.reply_count,
        'quote_count': tweet.quote_count,
        'user_id': tweet.user.id,
        'user_name': tweet.user.name,
        'followers_count': tweet.user.followers_count,
        'hashtags': tweet.hashtags,
        'place': tweet.place.full_name if tweet.place else 'N/A'
    })

```

```

tweets = await tweets.next()
await asyncio.sleep(5)

```

For each tweet, the following categories of metadata were collected:

- Tweet-level information (ID, timestamp, full text, language)
- Engagement metrics (likes, retweets, replies, quotes, views)
- User-level information (user ID, name, verification status, follower count)
- Entities (hashtags, URLs)
- Geographic information (when available)

Finally, the collected dataset was exported as a CSV file using UTF-8 encoding to preserve Tamil script integrity:

```

df = pd.DataFrame(tweets_data)
df.to_csv('tn_political_data_full_2.csv',
          index=False,
          encoding='utf-8-sig')

```

The resulting dataset formed the foundational corpus for manual annotation, preprocessing, and target-conditioned transformation.

Data Annotation Framework

This section describes the manual annotation strategy adopted to construct the supervised dataset for party mention detection. It includes political tweet filtering, multi-party labeling methodology, and the target-conditioned binary transformation applied to the annotated corpus.

Political Tweet Filtering

A total of 376 unique tweets were collected after merging the “Latest” and “Top” categories and removing duplicates. However, not all tweets retrieved through keyword-based querying were strictly political in nature.

Therefore, a manual filtering process was performed to identify genuinely political tweets. Each tweet was examined and assigned a binary label:

$$\text{is_political} \in \{\text{Yes}, \text{No}\}$$

Out of the 376 collected tweets:

- 229 tweets were manually annotated,
- Only tweets marked as Yes under is political were retained for further modelling.

This filtering step ensured that the final dataset exclusively represented political discourse and eliminated noise from non-political or ambiguous tweets.

Manual Annotation Process

The annotation process was conducted using Label Studio, an open-source data annotation platform that supports customizable labeling interfaces and multi-label classification tasks. Label Studio facilitated structured labeling, improved consistency, and enabled efficient management of annotation workflows.

After political filtering, each tweet was manually analyzed to determine whether it mentioned any of the following six political parties:

['ADMK', 'BJP', 'Congress', 'DMK', 'NTK', 'TVK']

Annotation was performed using the following criteria:

- Explicit textual mention of party names.
- Abbreviations in English (e.g., DMK, ADMK).
- Tamil script mentions (e.g., திமுக, அதிமுக).
- Contextual references clearly pointing to a specific party.

Each tweet could mention zero, one, or multiple parties. Therefore, the problem naturally exhibits a multi-label structure.

Target-Conditioned Binary Transformation

Instead of training a traditional multi-label classifier, a target-conditioned binary transformation was adopted.

Let:

N =Number of political tweets

$P=6$ (Number of political parties)

The dataset was expanded such that:

Total Samples= $N \times P$

For each political tweet, six training instances were generated — one for each party.

The transformation logic is shown below:

```
parties = ['ADMK', 'BJP', 'Congress', 'DMK', 'NTK', 'TVK']
party_mention_data = []

for idx, row in df.iterrows():
    if row['is_political'] == "Yes":
        for party in parties:
            is_mentioned = 1 if party in row['parties_mentioned'] else 0
            party_mention_data.append({
                'tweet_text': row['clean_text'],
                'Party': party,
                'is_mentioned': is_mentioned
            })

party_mention_df = pd.DataFrame(party_mention_data)
party_mention_df.to_csv(f'data/tweet_party_mention_{suffix}.csv',
                        index=False)
```

This transformation converts the original multi-label problem into multiple binary classification instances of the form:

$$f(\text{Tweet}, \text{Party}) \rightarrow \{0, 1\}$$

Where:

- 1 indicates that the given party is mentioned in the tweet,
- 0 indicates that the party is not mentioned.

This strategy is inspired by the Binary Relevance approach but enhanced through target conditioning, allowing the model to explicitly learn contextual associations between a tweet and a specific party.

The resulting dataset consists of the following fields:

- tweet text – Cleaned tweet content
- Party – Candidate political party
- is mentioned – Binary label

This structured transformation enables the use of transformer-based pairwise encoding of tweet-party combinations, forming the foundation for the TC-PED modeling framework.

Data Preprocessing Pipeline

This section describes the preprocessing steps applied to the annotated political tweet dataset prior to model training. Since social media text is inherently noisy, informal, and multilingual, a structured preprocessing pipeline was implemented to improve textual consistency while preserving semantic meaning.

Text Normalization

The first stage of preprocessing involved basic text normalization to ensure uniform representation across all tweets.

- All text was converted to lowercase.
- Unicode inconsistencies were standardized.

Lowercasing helps reduce vocabulary size and prevents the model from treating words such as “DMK” and “dmk” as separate tokens. This is particularly important in transformer-based architectures where token consistency affects embedding quality.

The lowercasing step was implemented as follows:

```
df['clean_text'] = df['full_text'].str.lower()
```

Emoji Handling (Demojization)

Tweets frequently contain emojis that convey sentiment, sarcasm, or emotional polarity. Instead of removing emojis directly, a demojization strategy was applied to preserve semantic information.

Demojization converts emojis into descriptive textual representations (e.g., 😊 → “smiling face”).

```
import emoji

df['clean_text'] = df['clean_text'].apply(
    lambda x: emoji.demojize(x))
```

This approach ensures that emotional cues remain embedded within the textual input and can be captured by the transformer model.

Noise Removal

Social media data often contains hyperlinks, mentions, and special characters that do not contribute directly to party entity detection. Therefore, the following cleaning steps were applied:

- URL removal
- Removal of unnecessary special characters

URL removal was performed using regular expressions:

```
import re

def remove_urls(text):
    return re.sub(r'http\S+|www\S+', "", text)

df['clean_text'] = df['clean_text'].apply(remove_urls)
```

This ensures that hyperlinks do not introduce irrelevant tokens into the model input.

Stop word Removal

Stop words are high-frequency functional words that typically carry limited semantic value in classification tasks. Since the dataset contains both Tamil and English tweets, stop word filtering was applied for both languages.

- English stop words were removed using standard NLP libraries.

- Tamil stop words were filtered using a predefined Tamil stop word list.

A simplified implementation for English stop word removal is shown below:

```
from nltk.corpus import stopwords

english_stopwords = set(stopwords.words('english'))

def remove_stopwords(text):
    tokens = text.split()
    filtered = [word for word in tokens
                if word not in english_stopwords]
    return " ".join(filtered)

df['clean_text'] = df['clean_text'].apply(remove_stopwords)
```

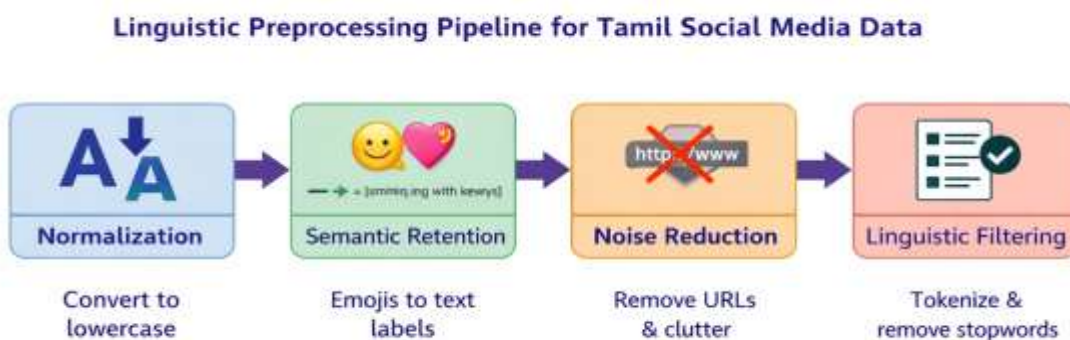
This step reduces noise while retaining informative tokens relevant to party mention detection.

Final Pre-processed Output

After applying all preprocessing steps, each tweet was represented in a cleaned textual format stored under the column:

clean_text

The preprocessing pipeline can be summarized as:



The cleaned text was then used to construct the target-conditioned dataset and subsequently tokenized for transformer-based modelling.

This structured preprocessing ensured improved textual consistency, reduced vocabulary sparsity, and enhanced model learning stability without eliminating critical semantic information.

Dataset Structure and Experimental Design

This section describes the final dataset structure after target-conditioned transformation and the experimental design adopted for training and evaluating the TC-PED model.

Final Dataset Structure

Following political filtering and target-conditioned binary transformation, the dataset was restructured into a pairwise format. Instead of treating party detection as a single multi-label task, each instance represents a *tweet-party pair*.

Each row in the final dataset consists of the following fields:

Field	Description
Tweet text	Pre-processed tweet text
Party	Candidate political party
is mentioned	Binary label (0 or 1)

Thus, each tweet contributes six training instances corresponding to the six political parties:

['ADMK', 'BJP', 'Congress', 'DMK', 'NTK', 'TVK']

Formally, for each political tweet T_i , the dataset contains:

$$(T_i, P_j) \rightarrow y_{ij}$$

where:

- P_j represents one of the six political parties,
- $y_{ij} \in \{0, 1\}$,
- $y_{ij}=1$ if party P_j is mentioned in tweet T_i ,
- $y_{ij}=0$ otherwise.

This structured expansion enables the modeling function:

$$f(\text{Tweet}, \text{Party}) \rightarrow \{0, 1\}$$

which forms the core of the Target-Conditioned Party Entity Detection (TC-PED) framework.

Train–Test Split Strategy

To evaluate model generalization, the dataset was divided into training and testing subsets using an 80:20 split ratio. Stratified sampling was applied based on the binary label is mentioned to preserve class distribution across splits.

The splitting procedure is shown below:

```
from sklearn.model_selection import train_test_split
train_df, test_df = train_test_split(
    df,
    test_size=0.2,
    random_state=42,
    stratify=df['is_mentioned']
)
```

The use of stratification ensures that:

- Both training and testing sets maintain proportional representation of positive and negative classes.
- Class imbalance effects are consistently reflected during evaluation.

The parameter `random_state = 42` was fixed to guarantee reproducibility of the experimental setup

Dataset Loader Implementation

To prepare the dataset for transformer-based training, a custom PyTorch dataset class was implemented. Each instance encodes the tweet text and the party name as a pair input to the tokenizer.

```
class TweetPartyDataset(Dataset):
    def __init__(self, dataframe, tokenizer, max_length=128):
        self.dataframe = dataframe.reset_index(drop=True)
        self.tokenizer = tokenizer
        self.max_length = max_length

    def __len__(self):
        return len(self.dataframe)
```

```

def __getitem__(self, idx):
    row = self.dataframe.iloc[idx]
    tweet = str(row['tweet_text'])
    party = str(row['Party'])
    label = row['is_mentioned']
    encoding = self.tokenizer(
        tweet,
        party,
        max_length=self.max_length,
        padding="max_length",
        truncation=True,
        return_tensors="pt"
    )

    return {
        'input_ids': encoding['input_ids'].squeeze(0),
        'attention_mask': encoding['attention_mask'].squeeze(0),
        'label': torch.tensor(label, dtype=torch.float)
    }

```

This implementation performs:

- Pairwise tokenization of tweet and party name,
- Padding and truncation to a maximum sequence length of 128 tokens,
- Conversion of labels to floating-point tensors for binary classification.

Finally, PyTorch DataLoaders were constructed for batch processing:

```

train_dataset = TweetPartyDataset(train_df, tokenizer)
test_dataset = TweetPartyDataset(test_df, tokenizer)

train_dataloader = DataLoader(train_dataset,
                               batch_size=4,
                               shuffle=True)

```

```
test_dataloader = DataLoader(test_dataset,
                             batch_size=4,
                             shuffle=False)
```

The batch size was set to 4 to accommodate GPU memory constraints while maintaining stable gradient updates.

This experimental design ensures:

- Reproducibility,
- Controlled evaluation,
- Efficient transformer-based training,
- Proper handling of class imbalance through stratified splitting.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the structural properties of the dataset before model development. The objective was to examine textual characteristics, identify class imbalance, and analyze the distribution of political and party-mention labels. These insights guided preprocessing decisions, model configuration, and metric selection.

Tweet Length Distribution

Tweet length (measured in characters) was analyzed to understand variability in textual input size. This is important for determining an appropriate maximum sequence length for transformer tokenization.

Histogram of Tweet Lengths

```
plt.figure(figsize=(10,4))
sns.histplot(df['tweet_len'].dropna(), bins=30, color='C2')
plt.title('Histogram of Tweet Lengths (characters)')
plt.xlabel('Tweet length (chars)')
plt.ylabel('Count')
plt.tight_layout()
plt.show()
df['tweet_len'].describe()
```

The histogram illustrates the overall distribution of tweet lengths. Most tweets cluster within a moderate range, while a smaller proportion of tweets appear significantly shorter or longer. The descriptive statistics (mean, median, quartiles) provide quantitative confirmation of central tendency and spread.

Understanding this distribution ensures that the chosen maximum token length captures most tweets without excessive padding.

Boxplot of Tweet Lengths

```
sns.boxplot(x=df['tweet_len'], color='C2')
plt.title('Boxplot of Tweet Lengths')
plt.xlabel('Tweet length (chars)')
plt.tight_layout()
plt.show()
```

The boxplot highlights:

- Median tweet length,
- Interquartile range (IQR),
- Presence of outliers.

Outliers represent unusually long or short tweets, which may influence token truncation decisions.

Political vs Non-Political Distribution

The political annotation dataset was examined to determine class balance between political and non-political tweets.

```
pol_path = 'data/tweet_is_political_2026-02-15.csv'
df_pol = pd.read_csv(pol_path, low_memory=False)
print('Political file rows:', len(df_pol))
df_pol.head()
```

Labels were normalized into two categories:

```
pol_label = df_pol['is_political'].astype(str).str.strip().str.lower()\
    .map(lambda x: 'Political' if x in ['yes','true','1']
        else 'Non-Political')
```

```
counts = pol_label.value_counts()
total = len(pol_label)
```

Pie Chart of Political Distribution

```
plt.figure(figsize=(6,6))
counts.plot.pie(autopct='%1.1f%%',
               colors=['C2','C0'],
               startangle=90,
               counterclock=False)
plt.title('Political vs Non-Political Tweets')
plt.ylabel("")
plt.tight_layout()
plt.show()
print('No. of political tweets:',
      int(counts.get('Political', 0)))
print('No. of Non-Political tweets:',
      int(counts.get('Non-Political', 0)))
print('Total No. of Tweets:', int(total))
```

This analysis reveals whether the dataset is balanced or skewed. If one class significantly dominates, it may bias the classifier toward the majority class. This observation supports the use of weighted loss functions and robust evaluation metrics such as MCC and F1-score.

Party Mention Distribution

The party mentioned dataset was analyzed to identify how frequently different political parties are referenced.

```
party_path = 'data/tweet_party_mention_2026-02-15.csv'
df_party = pd.read_csv(party_path, low_memory=False)
print('Party-mention file rows:', len(df_party))
df_party.head()
```

The `is_mentioned` column was cleaned and aggregated by party:

```

df_party['is_mentioned'] = pd.to_numeric(
    df_party['is_mentioned'],
    errors='coerce'
).fillna(0).astype(int)
party_counts = df_party.groupby('Party')['is_mentioned']\
    .sum()\
    .sort_values(ascending=False)
party_counts = party_counts[party_counts > 0]

```

Pie Chart of Party Mentions

```

plt.figure(figsize=(8,8))
wedges, texts = plt.pie(party_counts,
    startangle=90,
    counterclock=False)
plt.title('Party mentions in tweets')
plt.ylabel("")
plt.legend(labels=party_counts.index.tolist(),
    loc='center left',
    bbox_to_anchor=(1.0, 0.5))
plt.tight_layout()
plt.show()
total = int(party_counts.sum())
print('Party mention counts and percentages:')
for party, cnt in party_counts.items():
    pct = cnt / total * 100 if total > 0 else 0.0
    print(f'{party}: {int(cnt)} ( {pct:.1f}% )')
print('Total party mentions:', total)

```

This analysis helps identify:

- Which parties dominate the dataset,
- Whether mentions are evenly distributed,
- Potential representation bias.

Uneven party distribution may influence model learning and generalization.

EDA Insights

The exploratory analysis provides the following key insights:

- Tweet lengths are generally within a manageable range, supporting the selected maximum sequence length.
- The political classification task may exhibit class imbalance.
- Party mentions are not uniformly distributed, indicating possible representation bias.

These findings informed preprocessing choices, hyperparameter configuration, loss weighting strategy, and metric selection in the subsequent modeling phase.

TC-PED Model Architecture

This section describes the architectural design of the proposed Target-Conditioned Party Entity Detection (TC-PED) model. The framework leverages a pretrained transformer encoder combined with a custom classification head to determine whether a specific political party is mentioned within a given tweet.

Conceptual Framework

Traditional multi-label classification approaches attempt to predict all party mentions simultaneously. However, in this study, a *target-conditioned* formulation was adopted. Instead of predicting a multi-hot vector, the model learns a binary decision function:

$$f(\text{Tweet}, \text{Party}) \rightarrow \{0, 1\}$$

This means the model evaluates whether a particular party is mentioned in a given tweet by conditioning on the party name explicitly.

Such a formulation enables:

- Fine-grained contextual reasoning,
- Better handling of overlapping party mentions,
- Improved generalization across party entities.

Input Encoding Strategy

Each training instance consists of a pair:

$$(\text{Tweet Text}, \text{Party Name})$$

The input is tokenized using a transformer tokenizer in a pairwise format:

```
encoding = self.tokenizer(
    tweet,
    party,
    max_length=128,
    padding="max_length",
    truncation=True,
    return_tensors="pt"
)
```

Conceptually, the encoded input sequence follows the format:

[CLS]Tweet Text**[SEP]**Party Name**[SEP]**

where:

- **[CLS]** represents the classification token,
- **[SEP]** separates the tweet and party name.

The maximum sequence length was fixed at 128 tokens to balance contextual coverage and computational efficiency.

Base Encoder: Pretrained Transformer

The core encoder of the TC-PED model is a pretrained transformer model (IndicBERT-v2), which generates contextual embeddings for each token in the input sequence.

Given an input sequence of length L , the encoder produces:

$$\text{LastHiddenState} \in \mathbb{R}^{\text{batch} \times L \times 768}$$

where 768 denotes the hidden embedding dimension.

The encoder is fine-tuned during training, allowing it to adapt to the political entity detection task.

CLS Embedding Extraction

For classification tasks, the embedding corresponding to the **[CLS]** token is typically used as a global representation of the entire input sequence.

From the output tensor:

$$\mathbf{H} \in \mathbb{R}^{\text{batch} \times L \times 768}$$

the CLS embedding is extracted as:

$$\mathbf{h}_{\text{CLS}} = \mathbf{H}[:, \mathbf{0}, :]$$

where:

$$\mathbf{h}_{\text{CLS}} \in \mathbb{R}^{768}$$

This vector serves as the contextual summary representation of the tweet–party pair.

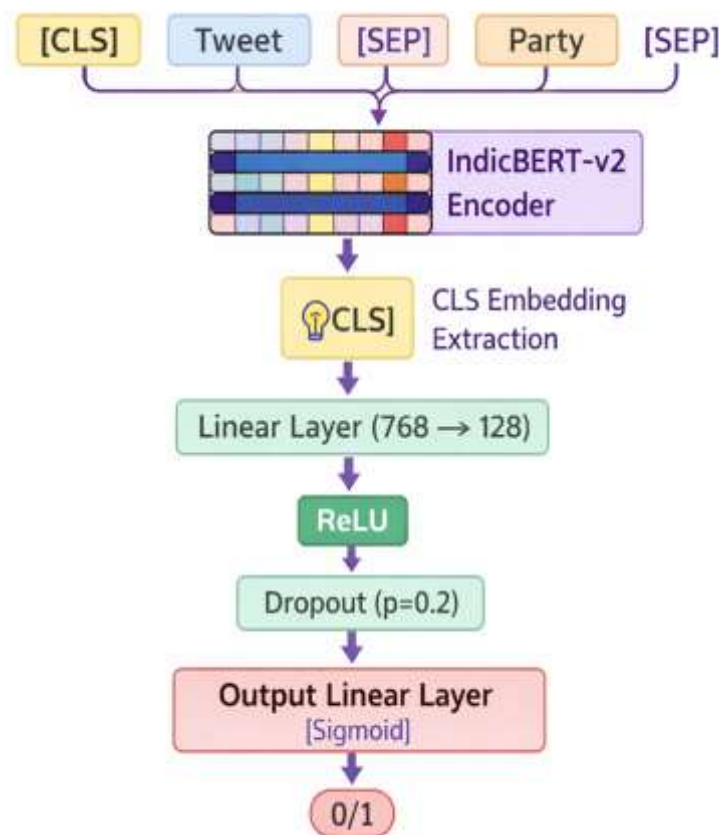
Classification Head

A custom feed-forward neural network was implemented on top of the CLS embedding to perform binary classification.

The architecture consists of:

1. Linear Layer: **768** $\rightarrow$ **128**
2. ReLU Activation
3. Dropout Layer (rate = 0.2)
4. Linear Layer: **128** $\rightarrow$ **1**

The model implementation is shown below:



```

class TCPEDModel(nn.Module):
    def __init__(self, base_model,
                  hidden_dim=768,
                  hidden_units=128,
                  dropout=0.2):
        super(TCPEDModel, self).__init__()
        self.base_model = base_model
        self.cls_linear = nn.Linear(hidden_dim, hidden_units)
        self.relu = nn.ReLU()
        self.dropout = nn.Dropout(dropout)
        self.classifier = nn.Linear(hidden_units, 1)
    def forward(self, input_ids, attention_mask):
        outputs = self.base_model(
            input_ids=input_ids,
            attention_mask=attention_mask
        )
        last_hidden_state = outputs.last_hidden_state
        cls_emb = last_hidden_state[:, 0, :]
        x = self.cls_linear(cls_emb)
        x = self.relu(x)
        x = self.dropout(x)
        logits = self.classifier(x)
        return logits.squeeze(-1)

```

Output Representation

The model outputs a single logit value:

$$\mathbf{z} \in \mathbb{R}$$

During evaluation, the sigmoid function is applied:

$$\hat{\mathbf{y}} = \sigma(\mathbf{z})$$

where:

$$\sigma(\mathbf{z}) = \frac{1}{1 + e^{-\mathbf{z}}}$$

A threshold of 0.5 is used to convert probabilities into binary predictions:

$$\hat{y} = \begin{cases} 1 & \text{if } \sigma(z) > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

Model Instantiation

The final model was instantiated as:

```
model = TCPEDModel(base_model,
                    hidden_units=128,
                    dropout=0.2)
```

This architecture effectively combines contextual semantic modeling from a pretrained transformer with a lightweight classification head tailored for target-conditioned political party detection.

Hyperparameter Configuration

This section presents the hyperparameter settings used for training the proposed TC-PED model. The hyperparameters were selected based on standard practices in transformer fine-tuning and were configured to ensure stable convergence, controlled regularization, and effective handling of class imbalance.

The complete set of hyperparameters used in the experiments is summarized in Table [1](#).

Hyperparameter Configuration for TC-PED Model

Parameter	Value
Learning Rate	2×10^{-5}
Weight Decay	0.01
Batch Size	4
Hidden Units (Classifier)	128
Dropout Rate	0.2
Maximum Sequence Length	128
Number of Epochs	30
Gradient Clipping Norm	1.0
Optimizer	AdamW

Parameter	Value
Loss Function	BCEWithLogitsLoss (Weighted)
Early Stopping Patience	5

Training Procedure

This section describes the complete training pipeline adopted for fine-tuning the TC-PED model. It includes the loss formulation, optimization strategy, gradient control mechanisms, and early stopping configuration used to ensure stable convergence and robust generalization.

Loss Function and Class Imbalance Handling

Since the task is formulated as a binary classification problem, the Binary Cross-Entropy with Logits Loss was used:

$$L = \text{BCEWithLogitsLoss}$$

Unlike standard binary cross-entropy, this formulation internally applies the sigmoid function, improving numerical stability.

Due to class imbalance in the dataset (fewer positive mentions compared to negative samples), a positive class weighting strategy was employed. The weight was computed as:

$$\text{pos_weight} = \frac{N_{\text{neg}}}{N_{\text{pos}}}$$

where:

- N_{neg} is the number of negative samples,
- N_{pos} is the number of positive samples.

The implementation is shown below:

```
num_pos = int(train_df['is_mentioned'].sum())
num_neg = max(1, len(train_df) - num_pos)
pos_weight = torch.tensor(
    [num_neg / (num_pos + 1e-9)],
    dtype=torch.float
).to(device)
loss_fn = nn.BCEWithLogitsLoss(pos_weight=pos_weight)
```

This weighting mechanism increases the contribution of minority-class samples to the loss, thereby reducing bias toward the majority class.

Optimization Strategy

The model parameters were optimized using the AdamW optimizer:

```
optimizer = AdamW(model.parameters(),  
                  lr=2e-5,  
                  weight_decay=0.01)
```

AdamW was selected due to its effectiveness in transformer fine-tuning tasks, particularly for handling large pre-trained models. The learning rate was fixed at 2×10^{-5} , which is commonly recommended for stable fine-tuning.

Gradient Clipping

To prevent gradient explosion and ensure stable training dynamics, gradient clipping was applied with a maximum norm of 1.0:

```
torch.nn.utils.clip_grad_norm_(  
    model.parameters(), 1.0  
)
```

Formally, gradient clipping enforces:

$$\| \mathbf{g} \| \leq 1.0$$

where $\mathbf{g}$ represents the gradient vector. This constraint stabilizes backpropagation, particularly in deep transformer architectures.

Early Stopping Mechanism

To prevent overfitting, an early stopping strategy was implemented with a patience value of 5 epochs. Training was terminated if the validation loss failed to improve for five consecutive epochs.

The early stopping class was defined as:

```
class EarlyStopping:  
    def __init__(self, patience=5, min_delta=0):  
        self.patience = patience  
        self.min_delta = min_delta  
        self.counter = 0
```

```

self.best_loss = None
self.early_stop = False
def __call__(self, val_loss):
    if self.best_loss is None:
        self.best_loss = val_loss
    elif val_loss < self.best_loss - self.min_delta:
        self.best_loss = val_loss
        self.counter = 0
    else:
        self.counter += 1
        if self.counter >= self.patience:
            self.early_stop = True

```

Whenever validation loss improved, the best-performing model checkpoint was saved:

```
torch.save(model.state_dict(), 'best_model.pth')
```

This ensured that the final evaluation was conducted using the model with optimal validation performance.

Training Loop

The model was trained for a maximum of 30 epochs. For each epoch, the following steps were performed:

1. Set model to training mode.
2. Iterate over batches from the training Data Loader.
3. Perform forward propagation.
4. Compute weighted binary loss.
5. Backpropagate gradients.
6. Apply gradient clipping.
7. Update parameters using the optimizer.

A simplified version of the training loop is shown below:

```

for epoch in range(epochs):
    model.train()
    train_loss = 0

```



```

for batch in train_dataloader:
    input_ids = batch['input_ids'].to(device)
    attention_mask = batch['attention_mask'].to(device)
    labels = batch['label'].to(device)
    optimizer.zero_grad()
    outputs = model(input_ids, attention_mask)
    loss = loss_fn(outputs, labels)
    loss.backward()
    torch.nn.utils.clip_grad_norm_(
        model.parameters(), 1.0
    )
    optimizer.step()
    train_loss += loss.item()

```

The average loss of training per epoch was computed and stored.

Testing and Evaluation Protocol

This section describes the procedure followed to evaluate the final TC-PED model on the held-out test set after training completion.

Model Checkpoint Loading

After training, the model checkpoint corresponding to the lowest validation loss was selected as the final model. This ensures that evaluation is performed using the best-performing validation model rather than the final training epoch.

The saved model weights were loaded as follows:

```

# Load best validation checkpoint
model.load_state_dict(torch.load('best_model.pth'))
model.to(device)
model.eval()

```

Evaluation was strictly conducted on the held-out test set, which was not used for parameter updates during training. This guarantees an unbiased estimation of model generalization performance.

Prediction Strategy

Since the model outputs raw logits, probabilities were obtained using the sigmoid activation function:

$$P(y=1|x)=\sigma(z)=\frac{1}{1+e^{-z}}$$

A fixed threshold of 0.5 was used to convert probabilities into binary predictions:

$$\hat{y}=\begin{cases} 1 & \text{if } \sigma(z) \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

Thus, the task is treated as a binary classification problem, where:

- 1 indicates party mention detected,
- 0 indicates no party mention.

Testing Loop

The complete evaluation loop used for generating predictions on the test set is shown below:

```
all_predictions = []
all_probabilities = []
all_labels = []
model.eval()
with torch.no_grad():
    for batch in test_dataloader:
        input_ids = batch['input_ids'].to(device)
        attention_mask = batch['attention_mask'].to(device)
        labels = batch['label'].to(device)
        # Forward pass
        logits = model(input_ids, attention_mask)
        # Convert logits to probabilities
        probabilities = torch.sigmoid(logits)
        # Apply threshold 0.5
        predictions = (probabilities >= 0.5).long()
        # Store results
        all_predictions.extend(predictions.cpu().numpy())
        all_probabilities.extend(probabilities.cpu().numpy())
        all_labels.extend(labels.cpu().numpy())
```

This testing loop ensures:

- No gradient computation during evaluation,
- Deterministic inference,
- Proper probability-to-binary conversion using a fixed threshold.

The resulting predictions and probabilities were then used to compute the final evaluation metrics.

Performance Metrics

To comprehensively evaluate the TC-PED model, multiple performance metrics were employed. Since the dataset exhibits class imbalance (i.e., the number of non-mention samples differs significantly from mention samples), relying solely on accuracy may lead to misleading conclusions.

Why Accuracy is Not Preferred

Accuracy is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

In imbalanced classification problems, a model can achieve high accuracy by simply predicting the majority of the time. For example, if 90% of samples belong to the negative class, a naive classifier predicting only negatives would achieve 90% accuracy while completely failing to detect positive cases.

Therefore, more robust metrics such as Precision, Recall, F1-score, and Matthews Correlation Coefficient (MCC) are preferred for reliable evaluation.

Precision

Precision measures the proportion of correctly predicted positive samples among all predicted positives:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Precision evaluates how reliable positive predictions are. A high precision indicates a low false positive rate.

Recall

Recall (Sensitivity) measures the proportion of correctly predicted positive samples among all actual positive samples:

$$\text{Recall} = \frac{TP}{TP + FN}$$

Recall reflects the model's ability to detect positive instances. High recall ensures that fewer true mentions are missed.

F1-Score

The F1-score is the harmonic mean of Precision and Recall:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

The harmonic mean penalizes extreme imbalances between precision and recall, making F1-score particularly suitable for imbalanced classification tasks.

Matthews Correlation Coefficient (Primary Metric)

The Matthews Correlation Coefficient (MCC) is selected as the primary evaluation metric because it provides a balanced measure even when classes are highly imbalanced.

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

MCC ranges from:

- +1: Perfect prediction
- 0: Random prediction
- -1: Complete disagreement

Unlike accuracy and F1-score, MCC incorporates all four components of the confusion matrix and provides a more informative and reliable performance measure in imbalanced settings.

ROC-AUC

The Receiver Operating Characteristic – Area Under Curve (ROC-AUC) measures the model's ability to rank positive samples higher than negative samples across all possible classification thresholds.

ROC-AUC is a threshold-independent metric, meaning it evaluates the discriminative power of the classifier without relying on a fixed decision threshold.

A higher ROC-AUC value indicates better separability between the positive and negative classes.

CHAPTER 4
DIAGRAMMATIC
REPRESENTATION
& STATISTICAL ANALYSIS

DIAGRAMMATIC REPRESENTATION & STATISTICAL ANALYSIS

INTRODUCTION

This chapter presents the statistical analysis and diagrammatic representation of the dataset and the performance evaluation of the proposed TC-PED model. The analysis includes descriptive statistics, distribution analysis, party-wise frequency analysis, training progression, and classification performance metrics.

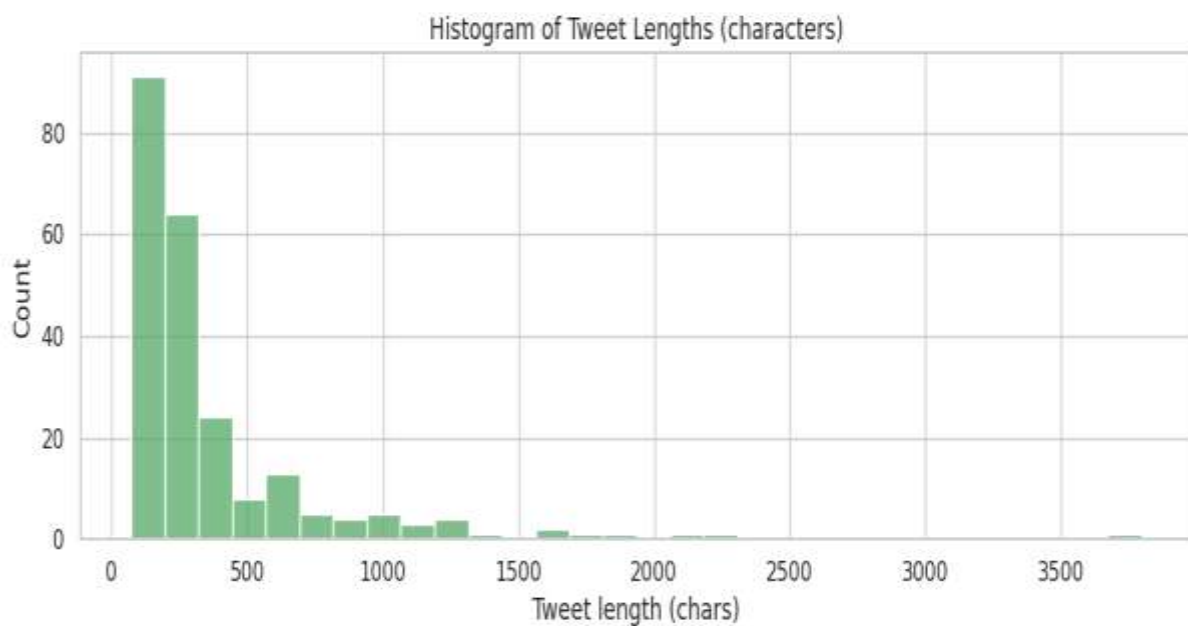
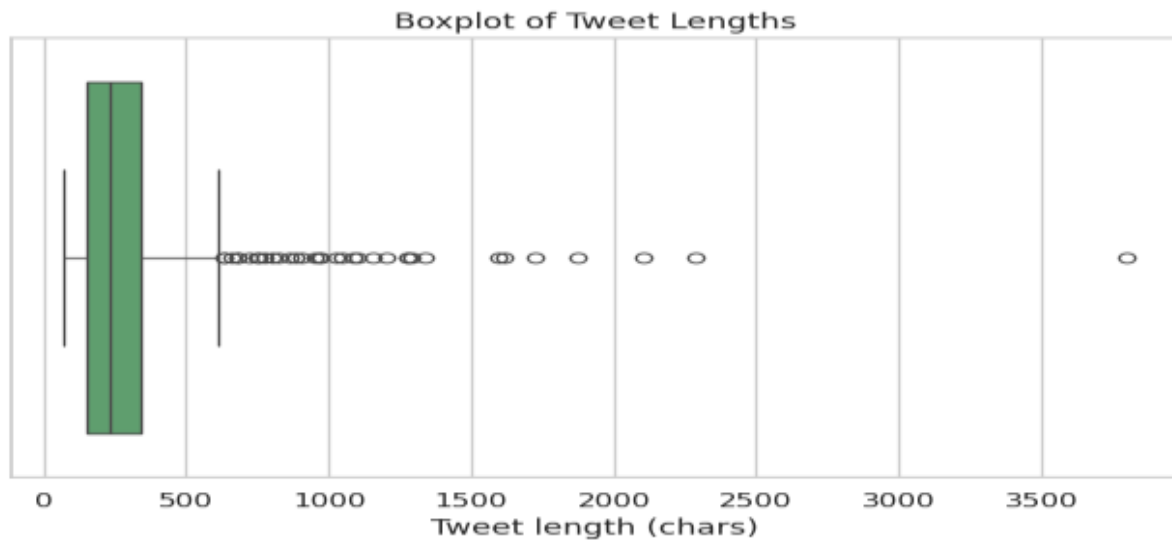
EXPLORATORY DATA ANALYSIS

1. DISTRIBUTION OF TWEET LENGTH

RESULT: Descriptive Statistics of Tweet Length

Statistic	Value
Count	229
Mean	378.23
Standard Deviation	427.97
Minimum	73
25th Percentile (Q1)	152
Median (50th Percentile)	231
75th Percentile (Q3)	338
Maximum	3796

DIAGRAM:



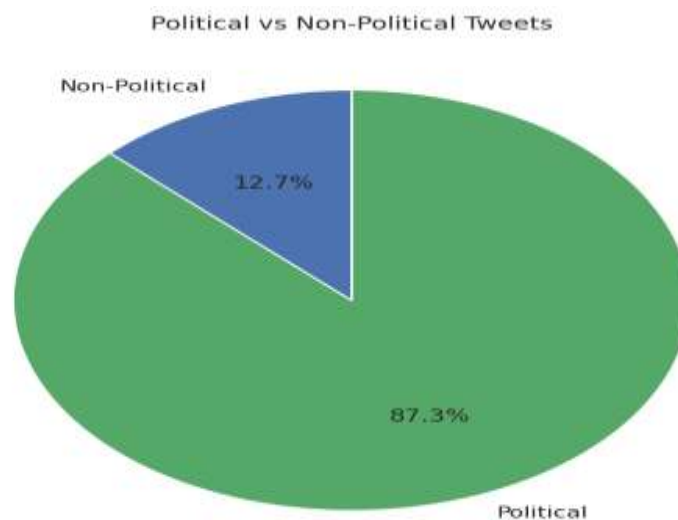
INTERPRETATION:

The descriptive statistics of tweet length clearly indicate that the distribution is positively skewed. The mean tweet length (378.23) is substantially higher than the median (231), showing that a small number of very long tweets are pulling the average upward. This right-skewed nature is further supported by the large standard deviation (427.97), which reflects high variability in the dataset. The maximum value of 3796 characters confirms the presence of extreme outliers, significantly distant from the central tendency. However, the quartile distribution reveals that 75% of the tweets are below 338 characters, meaning the

majority of observations are moderately sized and only a small proportion contributes to the long right tail. From a modelling perspective, this skewed and highly dispersed distribution justifies the implementation of transformer sequence truncation (`max_length = 128` tokens), as used in the TC-PED framework with IndicBERT-v2. Truncation ensures standardised input representation, computational stability, and efficient batching during training. Although extreme outliers may provide richer contextual information, they also increase computational complexity and memory requirements, thereby necessitating controlled sequence length for balanced model performance.

2. POLITICAL VS NON-POLITICAL TWEET DISTRIBUTION:

DIAGRAM:

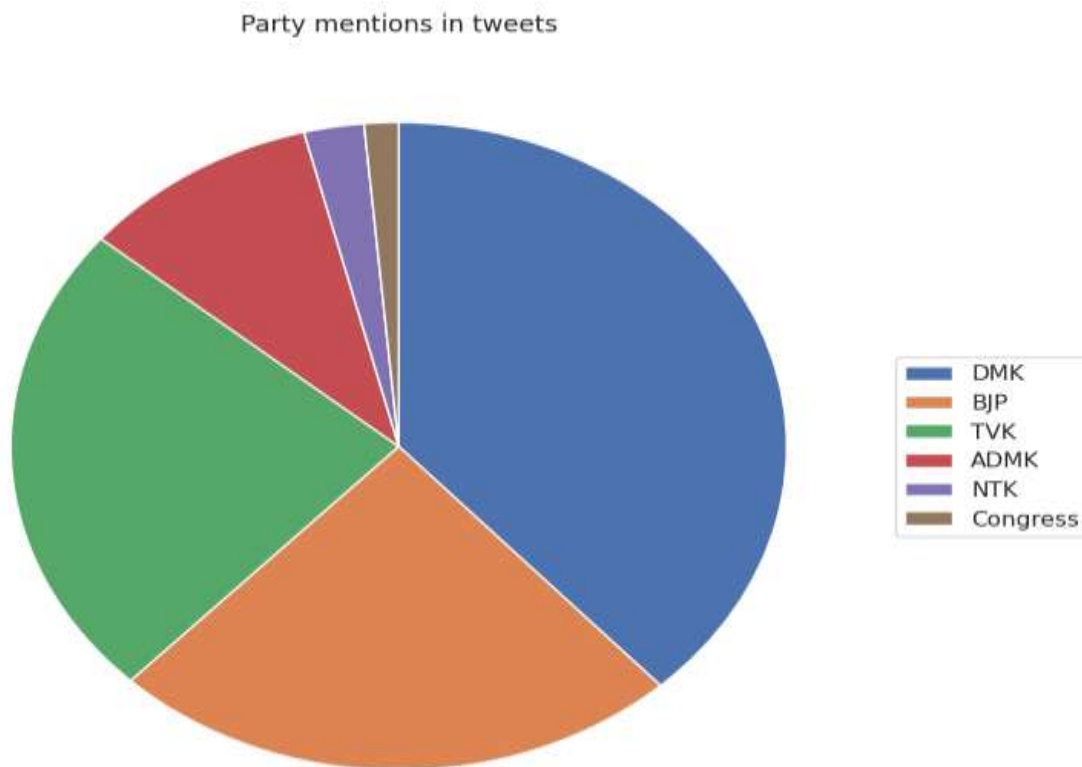


INTERPRETATION:

The pie chart illustrates that 87.3% of the collected tweets are political in nature, while only 12.7% are categorised as non-political. This distribution clearly indicates that the dataset is predominantly composed of political content. The high percentage of political tweets demonstrates that the data collection and filtering strategy effectively captured domain-specific political discourse relevant to the study. Since the primary objective of this study is to develop a Target-Conditioned Party Entity Detection (TC-PED) model for Tamil political tweets, the dominance of political content strongly supports and validates the research focus. At the same time, the presence of a smaller proportion of non-political tweets ensures necessary variability within the dataset, enabling the model to learn meaningful distinctions rather than assuming uniform political relevance across all tweets.

3. PARTY-WISE MENTION DISTRIBUTION:

DIAGRAM:



Party	Number of Mentions	Percentage (%)
DMK	107	38.2%
BJP	67	23.9%
TVK	67	23.9%
ADMK	28	10.0%
NTK	7	2.5%
Congress	4	1.4%
Total	280	100%

INTERPRETATION:

The party-wise mention distribution clearly shows that DMK dominates the political discourse in the dataset, accounting for 38.2% of the total mentions. This indicates that a significant portion of the collected tweets focus on DMK-related discussions, reflecting its

strong digital presence during the data collection period. BJP and TVK each contribute 23.9% of the mentions, demonstrating equal levels of online prominence and engagement within the sampled political conversations. In contrast, NTK (2.5%) and Congress (1.4%) are severely underrepresented, appearing only in a small fraction of the tweets. This uneven distribution across parties creates entity-level imbalance within the dataset, where some parties have substantially more training examples than others. As discussed in the modelling section of the project, such an imbalance justifies the use of a weighted Binary Cross-Entropy (BCE) loss function during training to prevent the model from becoming biased toward frequently mentioned parties and to ensure fair learning across all political entities.

MODEL TRAINING ANALYSIS:

RESULT:

Training Log Output:

Epoch 1/30, Train Loss: 1.1078, Val Loss: 1.0651

Best model saved at epoch 1

Epoch 2/30, Train Loss: 1.1130, Val Loss: 1.0499

Best model saved at epoch

Epoch 3/30, Train Loss: 1.2685, Val Loss: 1.0494

Best model saved at epoch 3

Epoch 4/30, Train Loss: 1.0141, Val Loss: 1.1705

Epoch 5/30, Train Loss: 0.9419, Val Loss: 0.9613

Best model saved at epoch 5

Epoch 6/30, Train Loss: 0.8206, Val Loss: 1.9352

Epoch 7/30, Train Loss: 0.7691, Val Loss: 1.8367

Epoch 8/30, Train Loss: 0.5827, Val Loss: 1.5830

Epoch 9/30, Train Loss: 0.5045, Val Loss: 1.6399

Epoch 10/30, Train Loss: 0.4708, Val Loss: 1.3432

Early stopping triggered

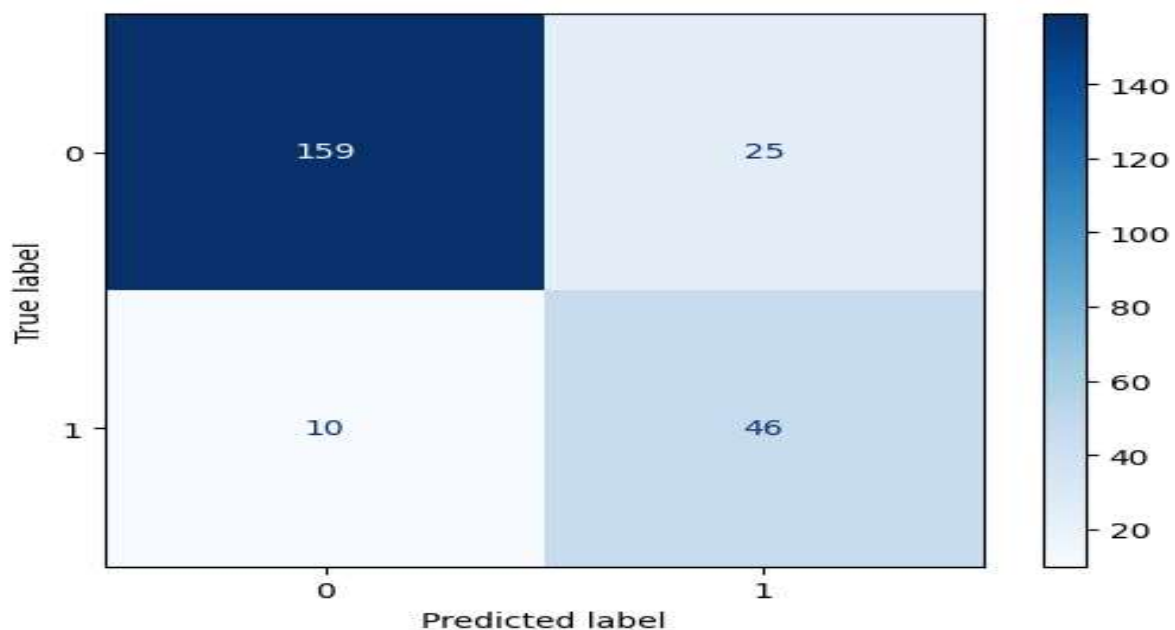
INTERPRETATION:

The epoch-wise loss progression shows that the validation loss steadily decreased until Epoch 5, reaching its minimum value of 0.9613. This indicates that the model was effectively learning meaningful patterns from the training data during the initial epochs. Accordingly, the best model checkpoint was saved at Epoch 5 for final evaluation. However, after this point, the validation loss increased sharply while the training loss continued to decrease, signalling the onset of overfitting. This divergence between training and validation curves suggests that the model began memorising the training data rather than generalising to unseen samples, which can be attributed in part to the relatively limited dataset size. The implementation of a patience-based early stopping mechanism successfully halted training at Epoch 10, preventing excessive overfitting. By selecting the optimal checkpoint at Epoch 5, early stopping enhanced the model's generalisation capability and ensured stable performance on the test set.

Model Performance Evaluation:

1. Confusion Matrix Analysis:

DIAGRAM:



INTERPRETATION:

The confusion matrix shows that the model achieved 159 True Negatives (TN) and 46 True Positives (TP), while producing 25 False Positives (FP) and 10 False Negatives (FN). This indicates that the classifier is highly effective in identifying non-mentions and correctly detecting a substantial number of actual mentions. The relatively low number of false negatives

reflects strong recall performance, meaning very few true mentions were missed. At the same time, the presence of 25 false positives suggests a moderate tendency to predict mentions in borderline cases. Overall, the distribution of TN, TP, FP, and FN demonstrates stable and reliable classification under imbalanced conditions.

OBSERVATIONS:

1. Model correctly identifies most non-mentions.
2. Only 10 actual mentions were missed
3. False positives (25) indicate a slight over-prediction tendency.
4. Strong sensitivity to actual positive cases.

2. PRECISION

RESULT:

Precision = 0.6479

INTERPRETATION:

The precision score of 64.79%, indicates that approximately two-thirds of the predicted mentions are correctly classified. The remaining incorrect positive predictions suggest that semantic overlaps or contextual similarities between parties may have contributed to false positives. This aligns with the model's evaluation profile, where recall is slightly prioritized over precision to ensure that most actual mentions are captured.

3. Recall

RESULT:

Recall = 0.8214

INTERPRETATION:

82.14% of actual mentions were successfully detected by the model, indicating high recall performance. This shows a very strong ability to capture entity presence within political tweets. Such performance is critical for political monitoring systems where missing true mentions can significantly affect analysis outcomes.

4. F1-Score:

RESULT:

F1-Score = 0.7244

INTERPRETATION:

The F1-Score represents the balanced harmonic mean of precision and recall. It indicates stable predictive behaviour by maintaining equilibrium between detecting true mentions and limiting false predictions.

5. Matthews Correlation Coefficient (Primary Metric):**RESULT:**

$$\text{MCC} = 0.6353$$

INTERPRETATION:

A strong correlation between predictions and actual labels is observed through the evaluation metric. It is suitable for imbalanced datasets and confirms that the model performs substantially better than a random classifier. This validates the robustness of the Target-Conditioned approach.

6. ROC-AUC:**RESULT:**

$$\text{ROC-AUC} = 0.8780$$

INTERPRETATION:

High discriminative ability is observed from the ROC-AUC performance. The model ranks positive cases effectively and indicates strong threshold-independent performance. This confirms classifier reliability across decision boundaries.

Overall Statistical Summary:**RESULT:**

- Highly skewed party distribution.
- Political tweets dominate the dataset.
- Model achieves:
- High Recall (82%)
- Strong ROC-AUC (0.878)
- Balanced F1-score (0.724)
- Robust MCC (0.63)

INTERPRETATION:

The TC-PED model demonstrates statistically reliable performance under real-world class imbalance, as reflected in the reported MCC of 0.6353 and ROC-AUC of 0.8780. The use of BCEWithLogitsLoss with a `pos_weight` parameter effectively addressed the significant imbalance between mentions and non-mentions. Early stopping preserved the model’s ability to handle unseen political discourse by preventing overfitting after the optimal validation loss at epoch 5. The confusion matrix results, along with strong recall and balanced F1-Score, indicate that the model captured contextual cues of Tamil political parties rather than defaulting to a “not mentioned” prediction. Overall, the results demonstrate the feasibility and robustness of the Target-Conditioned Party Entity Detection framework for Tamil political discourse.

CHAPTER 5

SUMMARY AND CONCLUSION

The project titled “**Target-Conditioned Party Entity Detection in Tamil Nadu Political Tweets**” has generated the following findings.

For this project, the total sample size consists of **229 annotated political tweets**, which were scraped from X (formerly Twitter) and manually labelled using Label Studio for six major political parties in Tamil Nadu.

FINDINGS:

- From the dataset summary, it is observed that the **average tweet length is 378 characters**, with maximum of 3796 characters, indicating high variability in tweet size and contextual richness.
- From the preprocessing and annotation process, it is inferred that **manual supervision was essential**, as political tweets contain sarcasm, indirect references, abbreviations, and contextual meanings.
- From the training results, the model achieved its **best validation loss (0.9613) at Epoch 5**, after which validation loss increased, indicating potential overfitting.
- Early stopping was triggered at **Epoch 10**, suggesting that the model converged quickly and did not benefit from prolonged training.
- From the comparative performance during training and validation, it is concluded the **IndicBERT embeddings performed effectively for Tamil political text representation**.
- From the Target-Conditioned modelling approach, it is inferred that conditioning the model on (Tweet, Party) pairs improves the ability to detect whether a specific party is mentioned in a tweet.
- The TC-PED approach demonstrates that **entity detection can be framed as a binary Classification problem using contextual embeddings**, rather than traditional Named Entity Recognition alone.
- The study reveals that **political data is highly dynamic**, and label distribution shifts frequently due to emerging parties, alliances, and changing narratives.
- From the experimental results, it is observed that **small dataset size limits generalisation**, and larger datasets are required for robust deployment.
- It is concluded that the **Target-Conditioned approach is not widely explored in political entity detection**, but this study provides empirical evidence of its potential.

CONCLUSION:

- This study demonstrates the feasibility of a **Target-Conditioned Political Entity Detection (TC-PED) model** for identifying party mentions in Tamil political tweets using IndicBERT embeddings.
- The findings indicate that **IndicBERT v2 performs well in capturing contextual semantics in Tamil political discourse**, making it suitable for low-resource regional language tasks.
- The Target-Conditioned approach provides a flexible framework where the same tweet can be evaluated against multiple political parties, improving fine-grained detection capability.
- However, due to the **small dataset size (229 tweets)**, the model requires further validation on large-scale datasets to establish robustness and scalability.
- Political discourse evolves rapidly, with **new parties emerging, alliances shifting, and narratives changing**, resulting in significant label shifts over time. Therefore, periodic retraining and continuous monitoring are essential.
- The study highlights that **manual supervision remains necessary**, as automated systems may fail to capture sarcasm, implicit references, and evolving political terminology.
- Overall, this study establishes that the Target-Conditioned framework has promising prospects in political text analytics, but its real-world deployment demands larger datasets, continuous updates, and domain-aware supervision.

SUGGESTIONS:

- Future study should expand the dataset to include **thousands of tweets across multiple election cycles** to improve generalisation.
- Implement cross-validation and temporal validation to handle **label shift and evolving political contexts**.
- Develop semi-supervised or active learning approaches to reduce the burden of manual annotation.
- Compare TC-PED with traditional Named Entity Recognition (NER) and multi-label classification models for performance benchmarking.
- Incorporate sentiment analysis alongside entity detection to provide deeper political insights.

- Establish a continuous retraining pipeline to handle **emerging political parties and changing** discourse patterns.
- Explore multilingual extensions by training similar models for other Indian regional languages.
- Integrate explainability techniques (such as attention visualization or SHAP analysis) to improve transparency in political AI systems.

BIBLIOGRAPHY

BIBLIOGRAPHY

JOURNALS:

- Khatavkar, V., Petkar, S., & Vaidya, A. S. (2025). Multilingual transformer contextual embedding model for political tweets analysis. *Cureus Journal of Computer Science*. <https://doi.org/10.7759/s44389-025-03177-4>
- Suriya, K. P., et al. (2025). Synapse@DravidianLangTech 2025: Multiclass political sentiment analysis in Tamil X (Twitter) comments: Leveraging feature fusion of IndicBERTv2 and lexical representations. In *Proceedings of the Fifth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages (NAACL 2025)*. <https://doi.org/10.18653/v1/2025.dravidianlangtech-1.122>
- Liu, W., & Cui, X. (2023). Improving named entity recognition for social media with data augmentation. *Applied Sciences*. <https://doi.org/10.3390/app13095360>
- Li, J., Sun, A., Han, J., & Li, C. (2020). A survey on deep learning for named entity recognition. *IEEE Transactions on Knowledge and Data Engineering*. <https://doi.org/10.1109/TKDE.2020.2981314>
- Nie, Y., Tian, Y., Wan, X., Song, Y., & Dai, B. (2020). Named entity recognition for social media texts with semantic augmentation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. <https://doi.org/10.18653/v1/2020.emnlp-main.107>
- Moon, S., Neves, L., & Carvalho, V. (2018). Multimodal named entity recognition for short social media posts. In *Proceedings of the North American Chapter of the Association for Computational Linguistics (NAACL)*. <https://doi.org/10.18653/v1/n18-1078>
- Derczynski, L., Nichols, E., Van Erp, M., & Limsopatham, N. (2017). Results of the WNUT2017 shared task on novel and emerging entity recognition. In *Proceedings of the Workshop on Noisy User-generated Text (WNUT@EMNLP)*. <https://doi.org/10.18653/v1/w17-4418>
- Lample, G., Ballesteros, M., Subramanian, S., Kawakami, K., & Dyer, C. (2016). Neural architectures for named entity recognition. In *Proceedings of the North American Chapter of the Association for Computational Linguistics (NAACL)*. <https://doi.org/10.18653/v1/n16-1030>

SOFTWARE:

- Python Programming Language
- PyTorch Deep Learning Framework
- IndicBERTv2 (AI4Bharat)
- Label Studio (Data Annotation Tool)

WEBSITES:

- <https://x.com>
- <https://huggingface.co/ai4bharat/indic-bert>
- <https://labelstud.io/>
- <https://pytorch.org/>

APPENDIX

DEPARTMENT OF STATISTICS

St. JOSEPH'S COLLEGE (AUTONOMOUS), TIRUCHIRAPPALLI-620 002

tweet_text	Party	is_mentioned
திருப்பூர் மாவட்டம் இடுவாய் கிராமத்தில் , பொதுமக்கள் விருப்பத்திற்கு எதிராக குப்பைக்கிடங்கு அமைக்கும் பணியை நிறுத்தக் கோரி , பொதுமக்களோடு இணைந்து ஆர்ப்பாட்டம் நடத்த , திமுக அரசு அனுமதி மறுத்ததோடு மட்டுமல்லாமல் , @ bjp4tamilnadu திருப்பூர் வடக்கு மாவட்டத் தலைவர் திரு @ kcmb_srinivasan மற்றும் நான் உட்பட பாஜக தொண்டர்களைக் கைது செய்திருக்கிறது காவல்துறை . நாட்டிலேயே , தூய்மையான பெருநகரங்களில் , சென்னை , மதுரை , கோவை மற்றும் சிறு நகரங்களில் ஈரோடு , திருப்பூர் உள்ளிட்டவை கடைசி இடங்களில் இருக்கின்றன . கமிஷன் வாங்குவதற்காக மட்டுமே , திருப்பூர் மாநகரத்தையே குப்பைக்கிடங்காக்கி வைத்திருக்கிறது திருப்பூர் மாநகராட்சி . திருப்பூர் மாநகர திமுக மேயர் , இதற்காக மட்டுமே பதவியில் இருக்கிறார் . பொதுமக்கள் எத்தனை போராட்டம் நடத்தினாலும் , திருப்பூருக்கான நிரந்தரத் தீர்வு , தேர்தலில் மக்கள் அளிக்கப் போகும் வாக்குகள்தான் . வரும் தேர்தலில் , இந்த மக்கள் விரோத திமுக அரசை வீட்டுக்கு	ADMK	0

அனுப்ப வேண்டும் . ஜனநாயக முறைப்படி , இடுவாய் கிராம பொதுமக்கள் பிரச்சினைக்காகக் குரல் கொடுக்க அனுமதி தர மறுத்து , எங்களைக் கைது செய்து அடக்குமுறையைக் கையாளும் திமுக அரசும் , காவல்துறையும் , நாளை முதல் வேறு வேறு வடிவங்களில் பொதுமக்களின் போராட்டத்தைச் சந்திப்பார்கள் என்பதைக் கூறிக் கொள்கிறேன் . திமுக அரசின் ஒடுக்குமுறைக்கும் கைது நடவடிக்கைகளுக்கும் என்றும் அஞ்சோம் . இடுவாய் மக்களின் குரலாக எங்கள் குரல் என்றும் ஒலிக்கும் . (1/2)		
திருப்பூர் மாவட்டம் இடுவாய் கிராமத்தில் , பொதுமக்கள் விருப்பத்திற்கு எதிராக குப்பைக்கிடங்கு அமைக்கும் பணியை நிறுத்தக் கோரி , பொதுமக்களோடு இணைந்து ஆர்ப்பாட்டம் நடத்த , திமுக அரசு அனுமதி மறுத்ததோடு மட்டுமல்லாமல் , @ bjp4tamilnadu திருப்பூர் வடக்கு மாவட்டத் தலைவர் திரு @ kcmb_srinivasan மற்றும் நான் உட்பட பாஜக தொண்டர்களைக் கைது செய்திருக்கிறது காவல்துறை . நாட்டிலேயே , தூய்மையான பெருநகரங்களில் , சென்னை , மதுரை , கோவை மற்றும் சிறு நகரங்களில் ஈரோடு , திருப்பூர் உள்ளிட்டவை கடைசி இடங்களில் இருக்கின்றன . கமிஷன் வாங்குவதற்காக	BJP	1

மட்டுமே , திருப்பூர் மாநகரத்தையே குப்பைக்கிடங்காக்கி வைத்திருக்கிறது திருப்பூர் மாநகராட்சி . திருப்பூர் மாநகர திமுக மேயர் , இதற்காக மட்டுமே பதவியில் இருக்கிறார் . பொதுமக்கள் எத்தனை போராட்டம் நடத்தினாலும் , திருப்பூருக்கான நிரந்தரத் தீர்வு , தேர்தலில் மக்கள் அளிக்கப் போகும் வாக்குகள்தான் . வரும் தேர்தலில் , இந்த மக்கள் விரோத திமுக அரசை வீட்டுக்கு அனுப்ப வேண்டும் . ஜனநாயக முறைப்படி , இடுவாய் கிராம பொதுமக்கள் பிரச்சினைக்காகக் குரல் கொடுக்க அனுமதி தர மறுத்து , எங்களைக் கைது செய்து அடக்குமுறையைக் கையாளும் திமுக அரசும் , காவல்துறையும் , நாளை முதல் வேறு வேறு வடிவங்களில் பொதுமக்களின் போராட்டத்தைச் சந்திப்பார்கள் என்பதைக் கூறிக் கொள்கிறேன் . திமுக அரசின் ஒடுக்குமுறைக்கும் கைது நடவடிக்கைகளுக்கும் என்றும் அஞ்சோம் . இடுவாய் மக்களின் குரலாக எங்கள் குரல் என்றும் ஒலிக்கும் . (1/2)		
திருப்பூர் மாவட்டம் இடுவாய் கிராமத்தில் , பொதுமக்கள் விருப்பத்திற்கு எதிராக குப்பைக்கிடங்கு அமைக்கும் பணியை நிறுத்தக் கோரி , பொதுமக்களோடு இணைந்து ஆர்ப்பாட்டம் நடத்த , திமுக அரசு அனுமதி	Congress	0

மறுத்ததோடு மட்டுமல்லாமல் , @ bjp4tamilnadu
 திருப்பூர் வடக்கு மாவட்டத் தலைவர் திரு @
 kcmb_srinivasan மற்றும் நான் உட்பட பாஜக
 தொண்டர்களைக் கைது செய்திருக்கிறது
 காவல்துறை . நாட்டிலேயே , தூய்மையான
 பெருநகரங்களில் , சென்னை , மதுரை ,
 கோவை மற்றும் சிறு நகரங்களில் ஈரோடு ,
 திருப்பூர் உள்ளிட்டவை கடைசி இடங்களில்
 இருக்கின்றன . கமிஷன் வாங்குவதற்காக
 மட்டுமே , திருப்பூர் மாநகரத்தையே
 குப்பைக்கிடங்காக்கி வைத்திருக்கிறது
 திருப்பூர் மாநகராட்சி . திருப்பூர் மாநகர திமுக
 மேயர் , இதற்காக மட்டுமே பதவியில்
 இருக்கிறார் . பொதுமக்கள் எத்தனை
 போராட்டம் நடத்தினாலும் , திருப்பூருக்கான
 நிரந்தரத் தீர்வு , தேர்தலில் மக்கள் அளிக்கப்
 போகும் வாக்குகள்தான் . வரும் தேர்தலில் ,
 இந்த மக்கள் விரோத திமுக அரசை வீட்டுக்கு
 அனுப்ப வேண்டும் . ஜனநாயக முறைப்படி ,
 இடுவாய் கிராம பொதுமக்கள்
 பிரச்சினைக்காகக் குரல் கொடுக்க அனுமதி தர
 மறுத்து , எங்களைக் கைது செய்து
 அடக்குமுறையைக் கையாளும் திமுக அரசும் ,
 காவல்துறையும் , நாளை முதல் வேறு வேறு
 வடிவங்களில் பொதுமக்களின்
 போராட்டத்தைச் சந்திப்பார்கள் என்பதைக்
 கூறிக் கொள்கிறேன் . திமுக அரசின்
 ஒடுக்குமுறைக்கும் கைது

நடவடிக்கைகளுக்கும் என்றும் அஞ்சோம் . இடுவாய் மக்களின் குரலாக எங்கள் குரல் என்றும் ஒலிக்கும் . (1/2)		
திருப்பூர் மாவட்டம் இடுவாய் கிராமத்தில் , பொதுமக்கள் விருப்பத்திற்கு எதிராக குப்பைக்கிடங்கு அமைக்கும் பணியை நிறுத்தக் கோரி , பொதுமக்களோடு இணைந்து ஆர்ப்பாட்டம் நடத்த , திமுக அரசு அனுமதி மறுத்ததோடு மட்டுமல்லாமல் , @ bjp4tamilnadu திருப்பூர் வடக்கு மாவட்டத் தலைவர் திரு @ kcmb_srinivasan மற்றும் நான் உட்பட பாஜக தொண்டர்களைக் கைது செய்திருக்கிறது காவல்துறை . நாட்டிலேயே , தூய்மையான பெருநகரங்களில் , சென்னை , மதுரை , கோவை மற்றும் சிறு நகரங்களில் ஈரோடு , திருப்பூர் உள்ளிட்டவை கடைசி இடங்களில் இருக்கின்றன . கமிஷன் வாங்குவதற்காக மட்டுமே , திருப்பூர் மாநகரத்தையே குப்பைக்கிடங்காக்கி வைத்திருக்கிறது திருப்பூர் மாநகராட்சி . திருப்பூர் மாநகர திமுக மேயர் , இதற்காக மட்டுமே பதவியில் இருக்கிறார் . பொதுமக்கள் எத்தனை போராட்டம் நடத்தினாலும் , திருப்பூருக்கான நிரந்தரத் தீர்வு , தேர்தலில் மக்கள் அளிக்கப் போகும் வாக்குகள்தான் . வரும் தேர்தலில் , இந்த மக்கள் விரோத திமுக அரசை வீட்டுக்கு அனுப்ப வேண்டும் . ஜனநாயக முறைப்படி ,	DMK	1

இடுவாய் கிராம பொதுமக்கள் பிரச்சினைக்காகக் குரல் கொடுக்க அனுமதி தர மறுத்து , எங்களைக் கைது செய்து அடக்குமுறையைக் கையாளும் திமுக அரசும் , காவல்துறையும் , நாளை முதல் வேறு வேறு வடிவங்களில் பொதுமக்களின் போராட்டத்தைச் சந்திப்பார்கள் என்பதைக் கூறிக் கொள்கிறேன் . திமுக அரசின் ஒடுக்குமுறைக்கும் கைது நடவடிக்கைகளுக்கும் என்றும் அஞ்சோம் . இடுவாய் மக்களின் குரலாக எங்கள் குரல் என்றும் ஒலிக்கும் . (1/2)		
திருப்பூர் மாவட்டம் இடுவாய் கிராமத்தில் , பொதுமக்கள் விருப்பத்திற்கு எதிராக குப்பைக்கிடங்கு அமைக்கும் பணியை நிறுத்தக் கோரி , பொதுமக்களோடு இணைந்து ஆர்ப்பாட்டம் நடத்த , திமுக அரசு அனுமதி மறுத்ததோடு மட்டுமல்லாமல் , @ bjp4tamilnadu திருப்பூர் வடக்கு மாவட்டத் தலைவர் திரு @ kcmb_srinivasan மற்றும் நான் உட்பட பாஜக தொண்டர்களைக் கைது செய்திருக்கிறது காவல்துறை . நாட்டிலேயே , தூய்மையான பெருநகரங்களில் , சென்னை , மதுரை , கோவை மற்றும் சிறு நகரங்களில் ஈரோடு , திருப்பூர் உள்ளிட்டவை கடைசி இடங்களில் இருக்கின்றன . கமிஷன் வாங்குவதற்காக மட்டுமே , திருப்பூர் மாநகரத்தையே	NTK	0

குப்பைக்கிடங்காக்கி வைத்திருக்கிறது திருப்பூர் மாநகராட்சி . திருப்பூர் மாநகர திமுக மேயர் , இதற்காக மட்டுமே பதவியில் இருக்கிறார் . பொதுமக்கள் எத்தனை போராட்டம் நடத்தினாலும் , திருப்பூருக்கான நிரந்தரத் தீர்வு , தேர்தலில் மக்கள் அளிக்கப் போகும் வாக்குகள்தான் . வரும் தேர்தலில் , இந்த மக்கள் விரோத திமுக அரசை வீட்டுக்கு அனுப்ப வேண்டும் . ஜனநாயக முறைப்படி , இடுவாய் கிராம பொதுமக்கள் பிரச்சினைக்காகக் குரல் கொடுக்க அனுமதி தர மறுத்து , எங்களைக் கைது செய்து அடக்குமுறையைக் கையாளும் திமுக அரசும் , காவல்துறையும் , நாளை முதல் வேறு வேறு வடிவங்களில் பொதுமக்களின் போராட்டத்தைச் சந்திப்பார்கள் என்பதைக் கூறிக் கொள்கிறேன் . திமுக அரசின் ஒடுக்குமுறைக்கும் கைது நடவடிக்கைகளுக்கும் என்றும் அஞ்சோம் . இடுவாய் மக்களின் குரலாக எங்கள் குரல் என்றும் ஒலிக்கும் . (1/2)		
திருப்பூர் மாவட்டம் இடுவாய் கிராமத்தில் , பொதுமக்கள் விருப்பத்திற்கு எதிராக குப்பைக்கிடங்கு அமைக்கும் பணியை நிறுத்தக் கோரி , பொதுமக்களோடு இணைந்து ஆர்ப்பாட்டம் நடத்த , திமுக அரசு அனுமதி மறுத்ததோடு மட்டுமல்லாமல் , @ bjp4tamilnadu	TVK	0

திருப்பூர் வடக்கு மாவட்டத் தலைவர் திரு @ kcmb_srinivasan மற்றும் நான் உட்பட பாஜக தொண்டர்களைக் கைது செய்திருக்கிறது காவல்துறை . நாட்டிலேயே , தூய்மையான பெருநகரங்களில் , சென்னை , மதுரை , கோவை மற்றும் சிறு நகரங்களில் ஈரோடு , திருப்பூர் மாநகர திமுக மேயர் , இதற்காக மட்டுமே பதவியில் இருக்கிறார் . பொதுமக்கள் எத்தனை போராட்டம் நடத்தினாலும் , திருப்பூருக்கான நிரந்தரத் தீர்வு , தேர்தலில் மக்கள் அளிக்கப் போகும் வாக்குகள்தான் . வரும் தேர்தலில் , இந்த மக்கள் விரோத திமுக அரசை வீட்டுக்கு அனுப்ப வேண்டும் . ஜனநாயக முறைப்படி , இடுவாய் கிராம பொதுமக்கள் பிரச்சினைக்காகக் குரல் கொடுக்க அனுமதி தர மறுத்து , எங்களைக் கைது செய்து அடக்குமுறையைக் கையாளும் திமுக அரசும் , காவல்துறையும் , நாளை முதல் வேறு வேறு வடிவங்களில் பொதுமக்களின் போராட்டத்தைச் சந்திப்பார்கள் என்பதைக் கூறிக் கொள்கிறேன் . திமுக அரசின் ஒடுக்குமுறைக்கும் கைது நடவடிக்கைகளுக்கும் என்றும் அஞ்சோம் . இடுவாய் மக்களின் குரலாக எங்கள் குரல் என்றும் ஒலிக்கும் . (1/2)		
thank erode : folded_hands_light_skin_tone :	ADMK	0
thank erode : folded_hands_light_skin_tone :	BJP	0

thank erode : folded_hands_light_skin_tone :	Congress	0
thank erode : folded_hands_light_skin_tone :	DMK	0
thank erode : folded_hands_light_skin_tone :	NTK	0
thank erode : folded_hands_light_skin_tone :	TVK	1
hon'ble pm thiru . @ narendramodi , tamil nadu backbone india ' export economy . punitive us tariffs hitting bedrock hard , disrupting production , orders livelihoods across industrial regions . urging immediate bilateral resolution protect tamil nadu ' exporters india ' global manufacturing strength .	ADMK	0
hon'ble pm thiru . @ narendramodi , tamil nadu backbone india ' export economy . punitive us tariffs hitting bedrock hard , disrupting production , orders livelihoods across industrial regions . urging immediate bilateral resolution protect tamil nadu ' exporters india ' global manufacturing strength .	BJP	1
hon'ble pm thiru . @ narendramodi , tamil nadu backbone india ' export economy . punitive us tariffs hitting bedrock hard , disrupting production , orders livelihoods across industrial regions . urging immediate bilateral resolution protect tamil nadu ' exporters india ' global manufacturing strength .	Congress	0
hon'ble pm thiru . @ narendramodi , tamil nadu backbone india ' export economy . punitive us tariffs hitting bedrock hard , disrupting production , orders livelihoods across industrial regions . urging immediate bilateral resolution protect tamil nadu ' exporters india ' global manufacturing strength .	DMK	0
hon'ble pm thiru . @ narendramodi , tamil nadu backbone	NTK	0

india ' export economy . punitive us tariffs hitting bedrock hard , disrupting production , orders livelihoods across industrial regions . urging immediate bilateral resolution protect tamil nadu ' exporters india ' global manufacturing strength .		
hon'ble pm thiru . @ narendramodi , tamil nadu backbone india ' export economy . punitive us tariffs hitting bedrock hard , disrupting production , orders livelihoods across industrial regions . urging immediate bilateral resolution protect tamil nadu ' exporters india ' global manufacturing strength .	TVK	0

Tweet Content
 @username
 Tweet Text

1. Was the tweet political?
☐ Yes^[1] ☐ No^[2]

2. Select the Political parties involved
☐ DMK^[3] ☐ ADMK^[4] ☐ TVK^[5] ☐ NTK^[6] ☐ BJP^[7] ☐ Congress^[8]